Immersive Studio Pro

User manual — editing for fulldome domes, flat screens and 360° rooms.

ALMA DIGITAL STUDIO

Version 1.0

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Contents

PART SIX · REFERENCE

PART ONE

Getting started

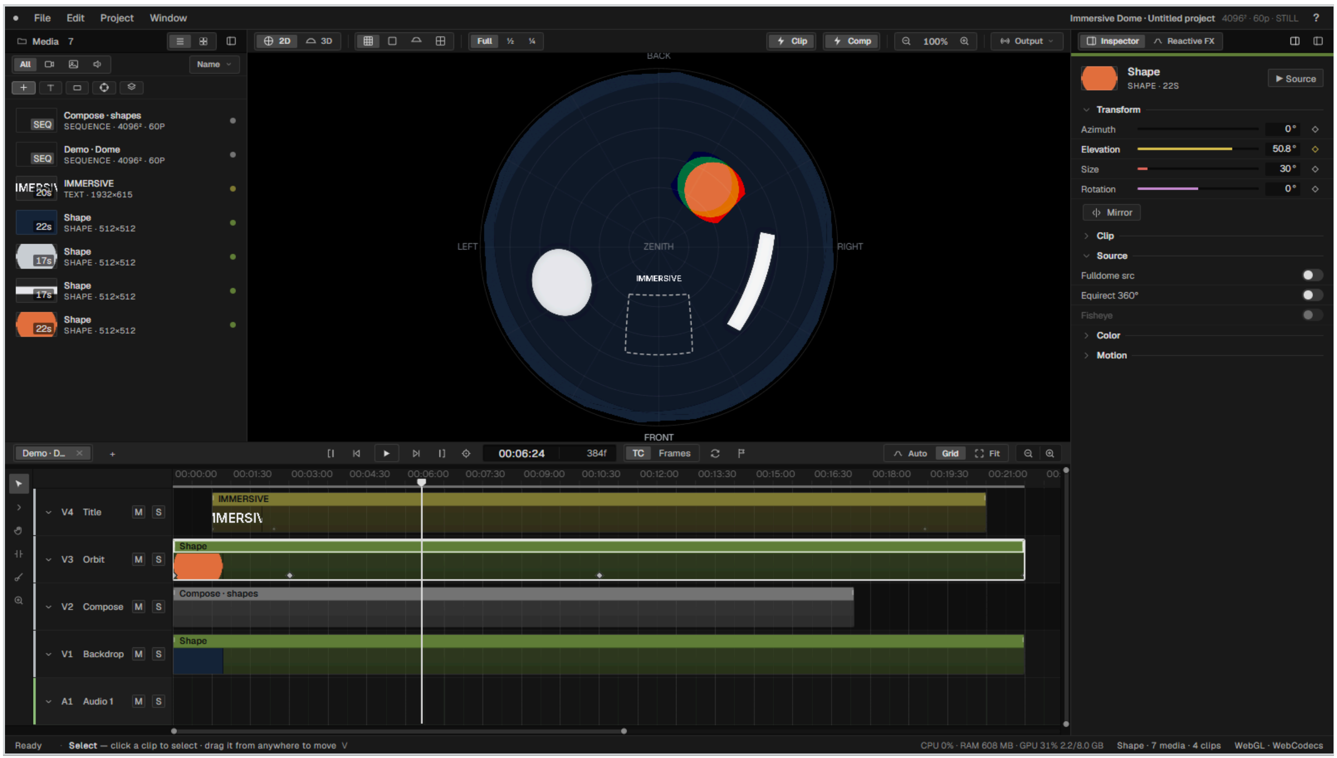
What the program is for, how to open it, and the handful of ideas that make everything else make sense. If you read only one part before you start working, read this one — immersive formats behave differently from a rectangular timeline, and the difference is worth ten minutes.

CHAPTER ONE

Welcome to Immersive Studio Pro

Immersive Studio Pro is a video editor for work that is not shown on a rectangle. It edits for fulldome domes, for ordinary flat screens, and for rooms whose walls and floor are all projection surfaces — and it treats all three as first-class formats rather than as an export option bolted onto a flat timeline.

It is a desktop application built by Alma Digital Studio for its own productions. Everything renders on the GPU through WebGL2: clips are warped into the dome in real time, effects run as shaders, and the master is composited to a single texture that the viewer, the live outputs and the exporter all read from. There is no render queue standing between you and what you are looking at.



The workspace in dome mode. Media on the left, the dome master in the centre, the inspector on the right, the timeline below. The circle is the whole of the projected image; nothing outside it is recorded.

What it is good at

Placing pictures on a dome, not on a plane

A clip in a dome sequence is positioned by azimuth, elevation and size — where it sits on the hemisphere — and the GPU warps it accordingly. You are steering a direction, not a rectangle.

Filling a dome with generated arrangements

The `Compose` tools build rings, grids, spirals, tunnels and woven basketry out of your own footage, as editable clips rather than as a black box.

Rooms with several surfaces

A 360° room is edited as one unrolled strip of walls plus a floor, previewed in 3D from the audience position, and exported whole or one wall at a time.

Delivering at real dome resolutions

4096 × 4096 H.264 and H.265 through the GPU encoder, lossless PNG sequences, and HAP for playback systems that want it.

What it is not

It is not a colour-grading suite, not an audio workstation, and not a general-purpose NLE competing with Premiere or Resolve on breadth. It carries the editing tools an immersive piece actually needs and stops there. Sound design happens elsewhere and comes back as a mix; heavy grading happens elsewhere and comes back as a LUT, which the program will apply per clip.

How this manual is arranged

Part one covers concepts and a first project. Part two walks the interface panel by panel. Parts three and four cover editing and the creative tools in the order you would meet them. Part five covers delivery. Part six is reference material — menus, shortcuts, parameter tables — meant for looking things up rather than reading.

Two conventions are used throughout. Interface labels appear as `Fulldome src`, exactly as they read on screen. Keys appear as `Ctrl`+`E`; on macOS read `Ctrl` as `⌘` unless stated otherwise.

> **THE SOFTWARE IS IN ENGLISH**
> The interface ships in English with a Spanish translation available. This manual documents the English labels.

CHAPTER TWO

Installing and launching

Immersive Studio Pro is a desktop application for Windows and macOS. It has no online component, no licence server and no project cloud: everything lives on your disk.

Requirements

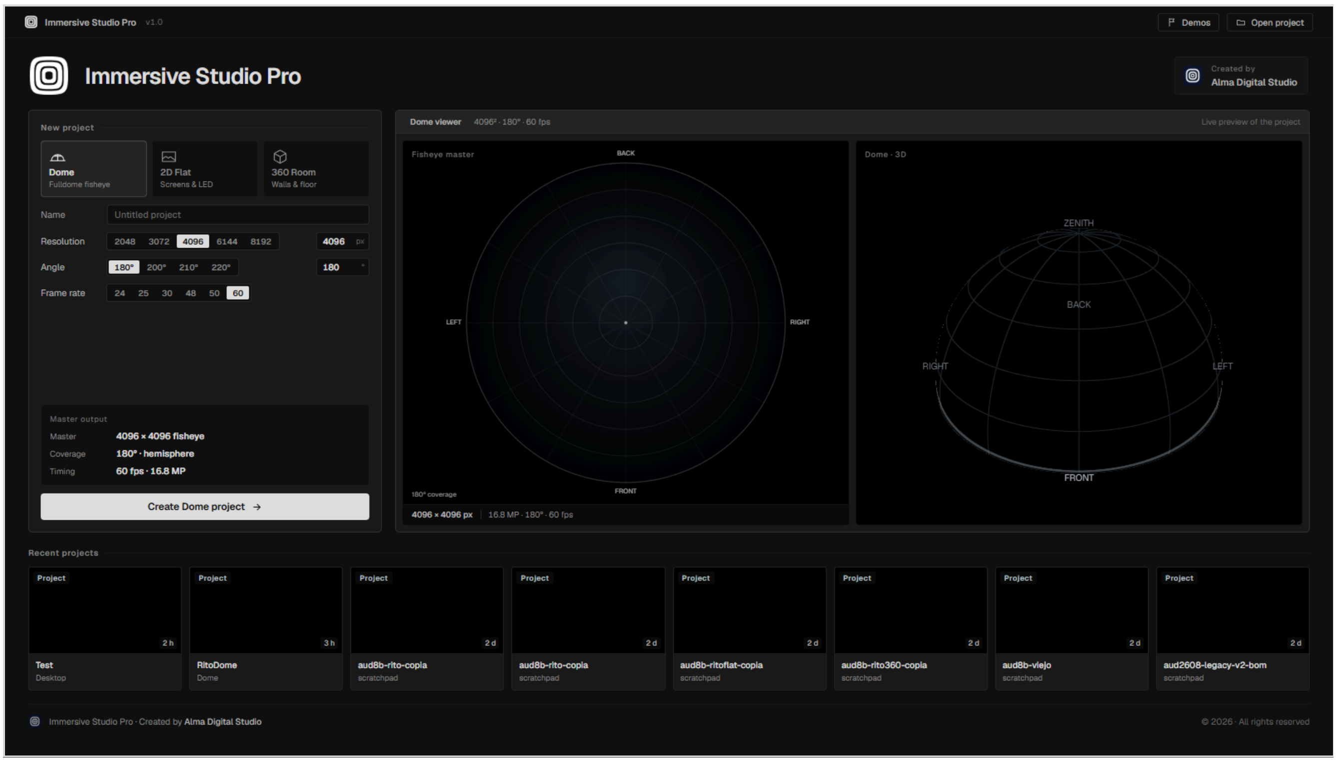
Windows	Windows 10 or 11, 64-bit. A discrete GPU is strongly recommended — the program asks the system for the high-performance adapter automatically on hybrid laptops.
macOS	Apple Silicon. Rendering and export both use the GPU.
Memory	16 GB is a working minimum for dome work at 4096^2. Decoding several layers of high-bitrate footage at once is the memory-hungry part, not the compositing.
Disk	Fast local storage for media. Exports at 4096^2 are large: budget roughly 300 MB per minute at 200 Mb/s.

Installing on Windows

Run the installer, Immersive Studio Pro Setup 1.0.0.exe. It installs per user, so it does not ask for administrator rights, and it registers the `.isp` project extension so that double-clicking a project opens it.

The start screen

The program opens on the start screen. This is where a project begins: you choose the format, set every parameter, and see a preview of the result before anything is created.



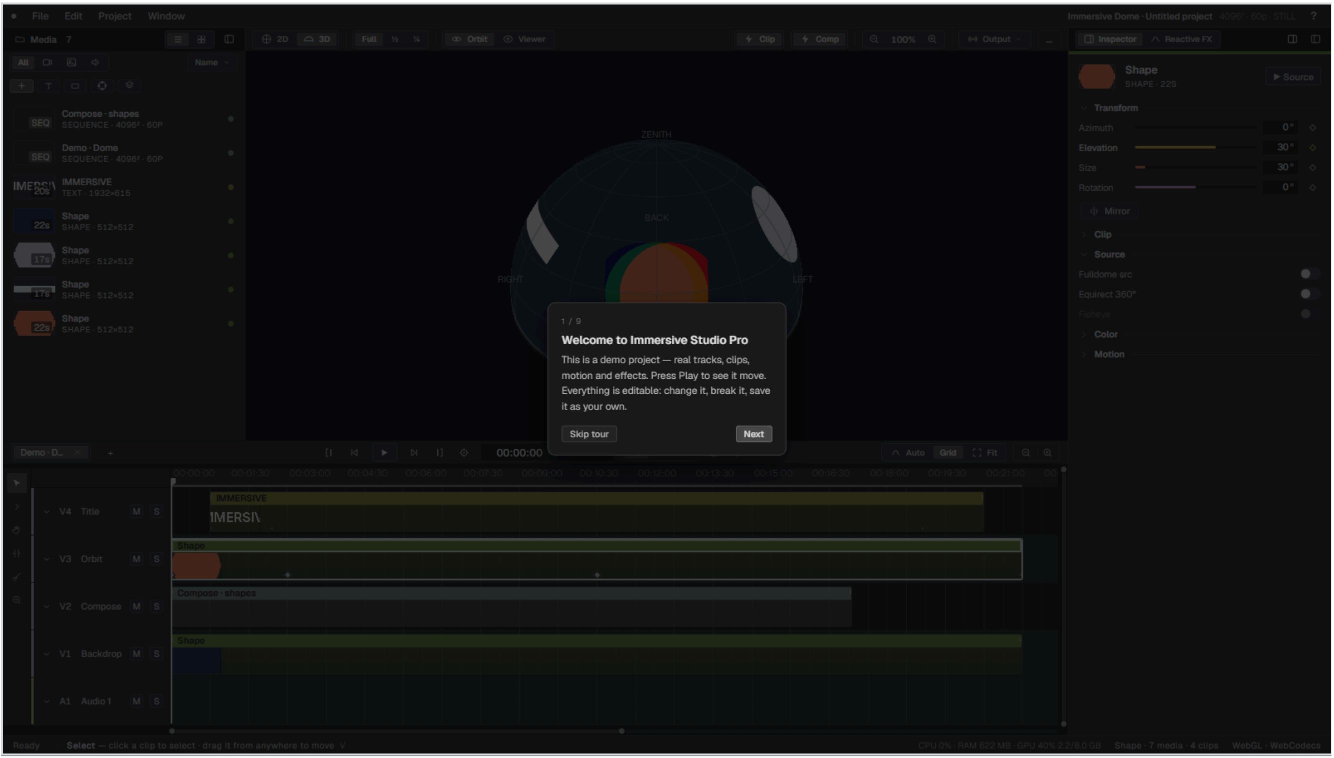
The start screen. Choose Dome, 2D or 360 Room; the panel on the right shows what you are about to make. `Demos` in the top bar builds a complete example project instead.

1. Pick a format: `Dome`, `2D` or `360 Room`.
2. Set the parameters — resolution, frame rate, dome coverage, or wall geometry for a room. The preview updates as you type.
3. Create the project. The editor opens with an empty timeline.

You can come back to this screen at any time with `File → New project…` (`Ctrl`+`N`). If a project is already open it stays open behind the screen, and `Back to project` returns to it untouched.

The demo projects and the guided tour

`Demos` builds a complete working project in the format you choose — several tracks, clips, a composition, motion, an effect and automation — and then runs a short guided tour over it. Nothing in it is locked: change it, break it, save it as your own. It is the fastest way to see how the pieces fit.



The guided tour highlights one region of the interface at a time. `→` and `←` move through it, `Esc` ends it. You can relaunch it from `Window → Guided tour`.

Preferences

`Project → Preferences…` holds the application-wide settings, including the interface language. Project settings — resolution, frame rate, dome coverage — live in `Project → Sequence settings…` instead, because they belong to the sequence rather than to the program.

CHAPTER THREE

Immersive concepts

Six ideas explain most of the program's behaviour. They are worth understanding before you start, because several of them have no equivalent in a flat editor and guessing produces frustrating results.

The three formats

Every sequence is in one of three modes, chosen when it is created and shown in the format chip at the top right of the window.

Dome
: A fulldome image: a circular fisheye covering a hemisphere. Clips are placed by direction — azimuth and elevation — and warped onto the dome by the GPU. The master is square, and only the inscribed circle is projected.

2D
: An ordinary rectangular canvas. Clips are placed by x, y and scale. Useful on its own for flat-screen work, and used internally as the workspace for compositions that later enter a dome.

360 Room
: A room whose walls are projection surfaces, plus an optional floor. The walls are unrolled into one wide rectangular strip that you edit like a 2D canvas; a 3D view shows what the audience will see.

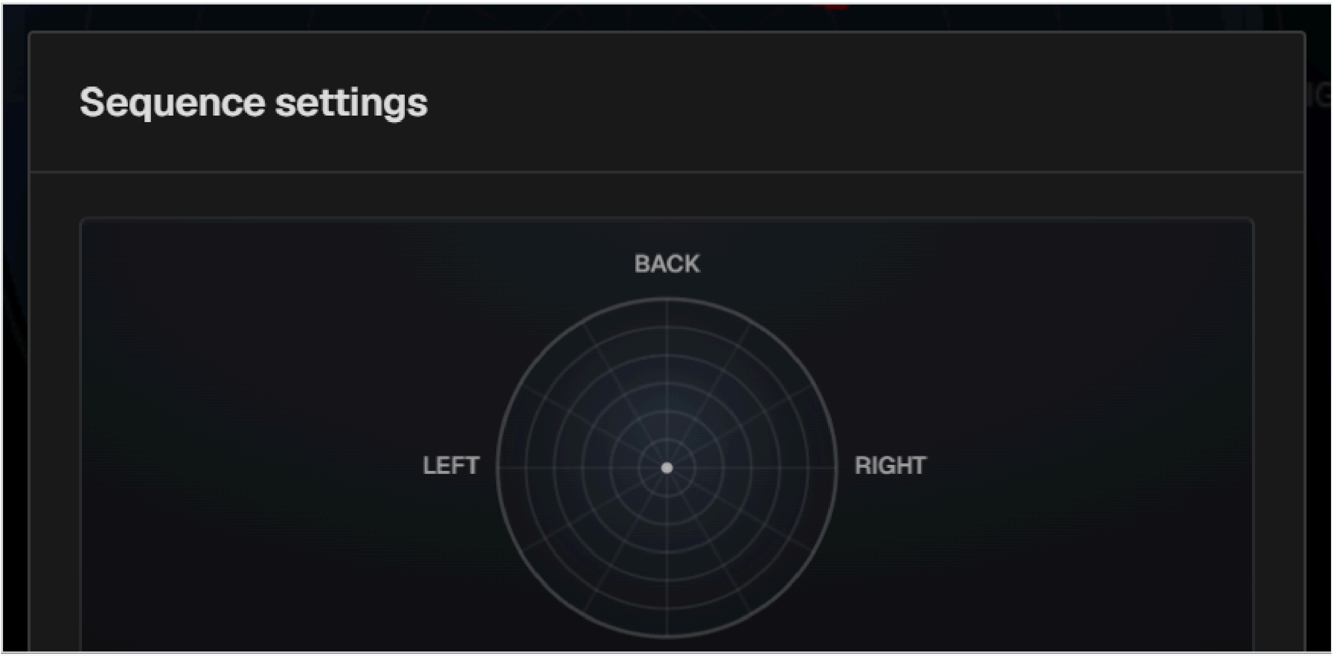
The composite master

Whatever the format, every visible clip is drawn into a single master image each frame. The viewer shows that master; NDI and Spout broadcast it; the exporter records it. There is no separate "render" pass with different rules, which is why what you see while editing is what you get on delivery.

In a dome sequence the master is square and the image is a fisheye circle inside it. **Nothing outside that circle is part of the picture.** The corners are black, or transparent if you export with alpha. This is not a crop applied at the end — it is what a fulldome master is.

Dome coverage

A fulldome master is an azimuthal-equidistant fisheye, and its *coverage* is the field of view it represents: 180° for a true hemisphere, or 200°, 210° or 220° for domes that continue below the horizon. The setting lives in `Project → Sequence settings…` and changes the projection everywhere at once — the warp, the 2D guides, the 3D preview and the export.



Sequence settings. Resolution, frame rate and dome coverage. Coverage updates live, so you can watch the framing change as you choose.

> **MATCH THE DOME YOU ARE DELIVERING TO**
> Coverage is not a look, it is a statement about the venue. A piece mastered at 180° and projected in a 210° dome will be wrong at the edges, and no amount of repositioning fixes it. Set it once, at the start, from the venue's specification.

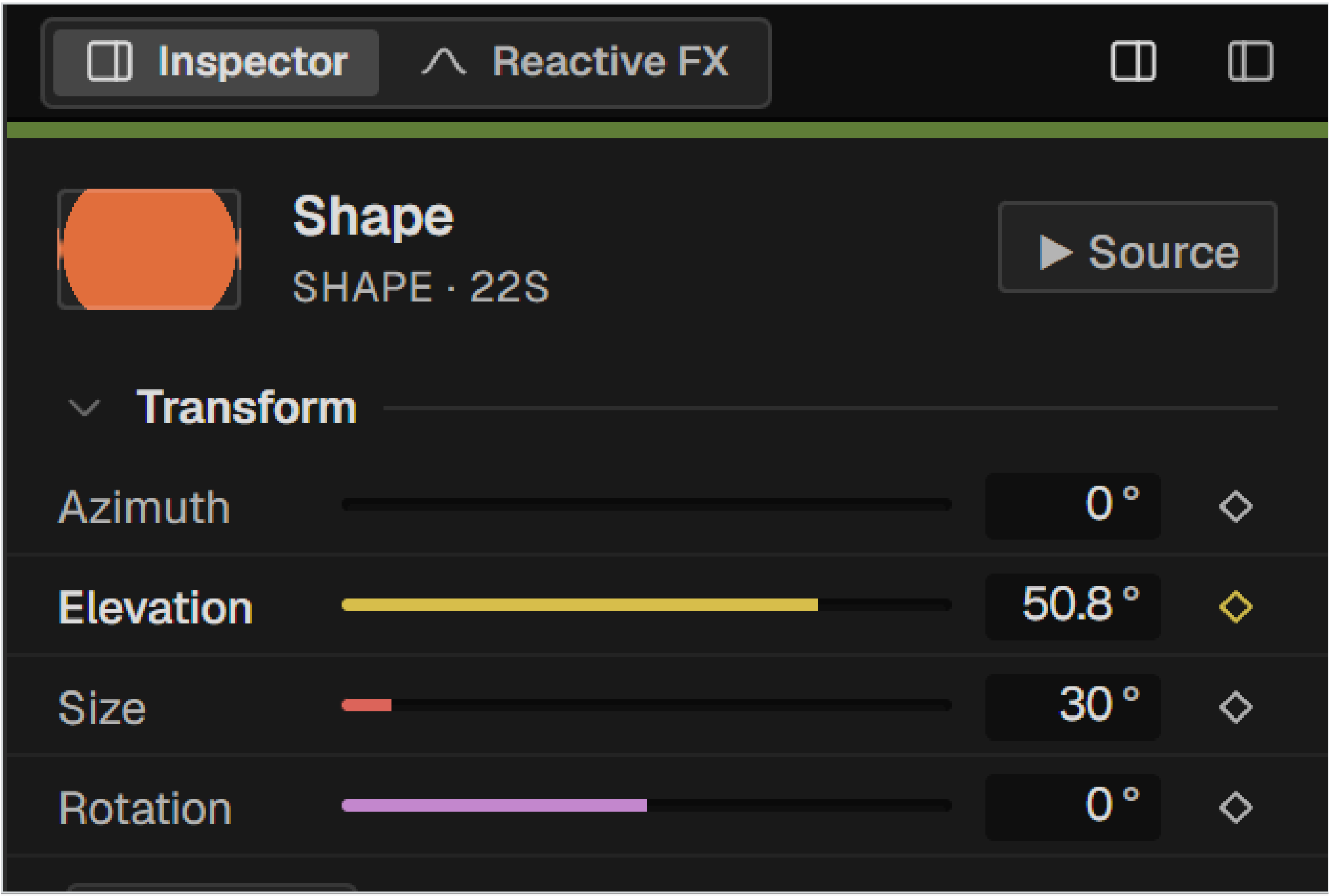
Fulldome source versus a placed clip

This is the distinction that causes the most confusion, and it is simple once stated. A clip in a dome sequence is treated in one of two ways:

A placed clip (`Fulldome src` off)
: The footage is an ordinary rectangular picture. It is placed somewhere on the dome — a direction and a size — and the GPU bends it to sit on the curved surface. This is what you want for material that was not shot for a dome: a face, a title, a texture.

A fulldome source (`Fulldome src` on)
: The footage *is already* a dome master — a circular fisheye filling the frame. It is drawn straight into the master one-to-one, not placed anywhere, because it already covers the whole sky. Compositions generated by the program are fulldome sources by nature.

A third switch, `Equirect 360°`, handles equirectangular 360° footage and reprojects it into the dome. `Fisheye` — with its own `Amount` — bends a fulldome source further, and is only available when `Fulldome src` is on, because bending a picture that is not covering the dome has no meaning.



Transform and Source in the inspector. Azimuth, elevation, size and rotation position a placed clip; the three switches below decide how the footage is interpreted in the first place.

Sequences are media

A sequence is not a separate document — it is an item in the media pool, and it can be dropped into another sequence as a clip. That single idea gives you Premiere-style tabs, nesting, and generated compositions, all with the same machinery. Chapter 15 covers it properly.

Everything is a clip

Video, stills, audio, text, shapes, generated compositions, nested sequences and adjustment layers are all clips on tracks. They trim the same way, they carry the same inspector sections where those apply, and they animate the same way. There is no separate title tool or effects layer with its own rules.

CHAPTER FOUR

Your first project

A short walkthrough that touches the main parts of the program: create a dome sequence, bring in footage, place it on the dome, animate it, add a generated composition and export a master. Twenty minutes, start to finish.

1 — Create the sequence

1. On the start screen choose `Dome`.
2. Set the resolution to **4096 × 4096** and the frame rate to the venue's — 30 or 60.
3. Set **coverage** to whatever the dome is: 180° unless told otherwise.
4. Create the project.

2 — Bring in footage

Drag files onto the media panel, or press Ctrl+I. Video, stills, audio and image sequences are all accepted. If a clip is heavy, the status bar will suggest generating a proxy — right-click the item and choose `Generate proxy`. Proxies are never made automatically; see chapter 10.

3 — Put a clip on the dome

1. Drag a clip from the media panel to a video track.
2. Select it. In the inspector, open `Transform`.
3. Drag `Azimuth` to swing it around the horizon, `Elevation` to raise it towards the zenith, and `Size` to make it bigger.
4. Or simply drag the clip in the viewer — the same values move.

> **ELEVATION GOES BELOW ZERO**
> Negative elevation places a clip below the horizon. In a dome with coverage above 180° that is real projected area, not a mistake.

4 — Make it move

Open `Motion` in the inspector and click a preset — `Orbit` makes it travel around the horizon, `Float` gives it a slow untethered drift. Presets are not black boxes: each one drops ordinary modifiers into the list, and you can retune, disable or delete them one at a time.

For a movement with an exact start and end, use keyframes instead: put the playhead where you want it, set the value, and the program records a keyframe automatically — the same rule as After Effects. Chapter 19 covers both.

5 — Fill the dome with a composition

1. Select two or three media items in the panel.
2. Click `Compose` in the create row.
3. Choose a type — `Ring` to start with — and set the count.
4. Confirm. A composition appears on the timeline as a single clip.

The composition is a nested sequence with real clips inside it. Double-click it to go in and edit them by hand; the parameters that generated it stay available in the inspector.

6 — Export a master

1. `File → Export...` (Ctrl+Shift+E).
2. Choose the destination folder with `Choose...` — before you start, not after.
3. Pick a codec. `MP4 H.264 · GPU, up to 4096` is the fast path for a 4K dome master.
4. Set `Quality` to `Maximum` and check the bitrate.
5. Press `Export`.

Save your project as you go with Ctrl+S. Projects are `.isp` files, written atomically with a `.bak` alongside.

PART TWO

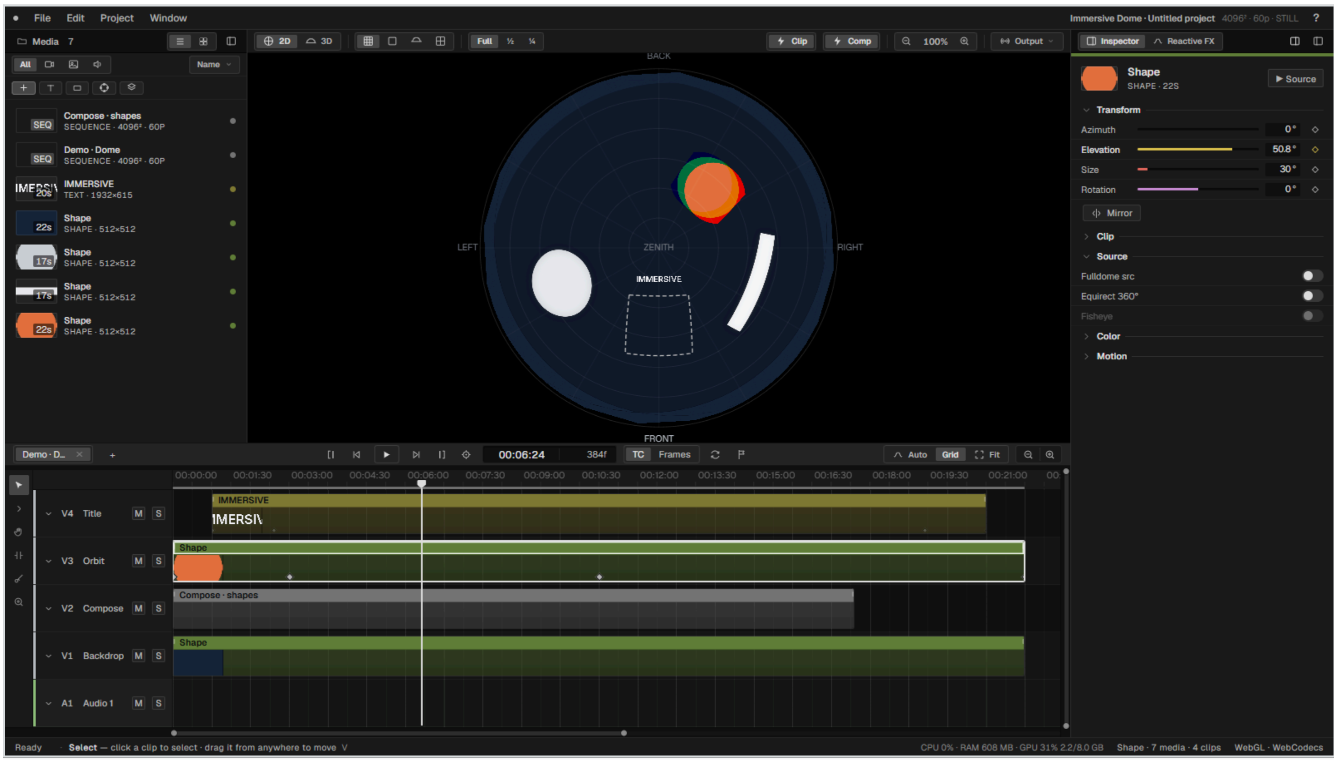
The workspace

A tour of the four panels and the bars that surround them. This part describes what each control does; the parts that follow describe what to do with them.

CHAPTER FIVE

The workspace at a glance

One window, four panels, and no floating clutter. The layout does not change between formats — only what the viewer draws changes.



The whole window. Menu bar and format chip on top; media, viewer and inspector across the middle; transport and timeline below; status bar at the foot.

The regions

Menu bar

Left of the top bar: `File`, `Edit`, `Project`, `Window`. Chapter 25 lists every entry.

Format chip

Top right. Shows the active sequence's format, resolution, frame rate and export codec. Click it to open the sequence settings.

Media panel

Left. Everything the project knows about: footage, stills, audio, text, shapes, sequences and compositions.

Viewer

Centre. The master image, in 2D or in 3D preview.

Inspector

Right. Everything about the selected clip, in collapsible sections. A second tab, `Reactive FX`, holds the audio-driven effects chain.

Transport
Between the viewer and the timeline: playback controls, timecode, sequence tabs and the editing well.

Timeline
Bottom. Tracks, clips, ruler, markers and the automation lanes.

Tool rail
The narrow strip at the left edge of the timeline: the six editing tools.

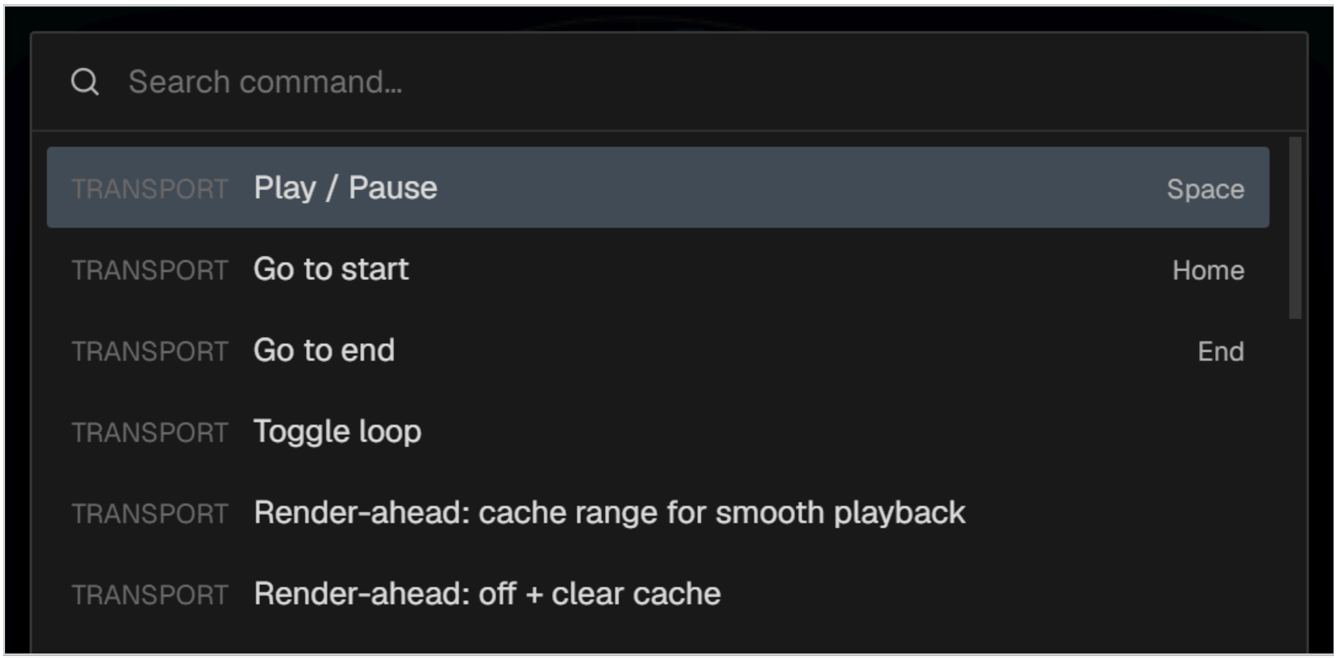
Status bar
The foot of the window. Left: what the current tool does. Right: CPU, memory and GPU load, and the project's media and clip counts.

Resizing and hiding panels

Drag the gutters between panels to resize them. The media and inspector panels can be collapsed entirely from `Window`, or with the two small buttons at the top right of the inspector. The timeline's height is dragged from its top edge.

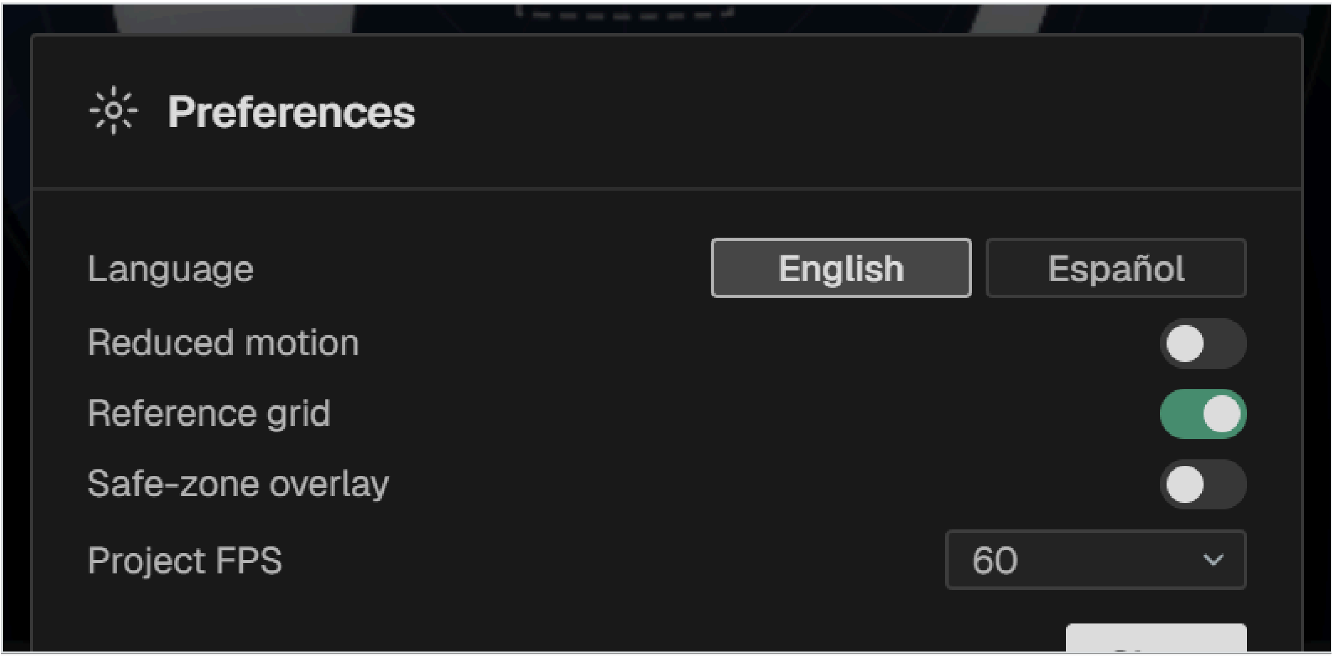
The command palette

Ctrl+K, F1 or ? opens a searchable list of every command in the program with its shortcut. It is the fastest way to find something whose menu you have forgotten, and the fastest way to learn the shortcuts.



The command palette. Type to filter, arrows to move, Enter to run. The right-hand column shows the shortcut where one exists.

Preferences



Preferences holds the application-wide settings, including the interface language.

Full performance and the second window

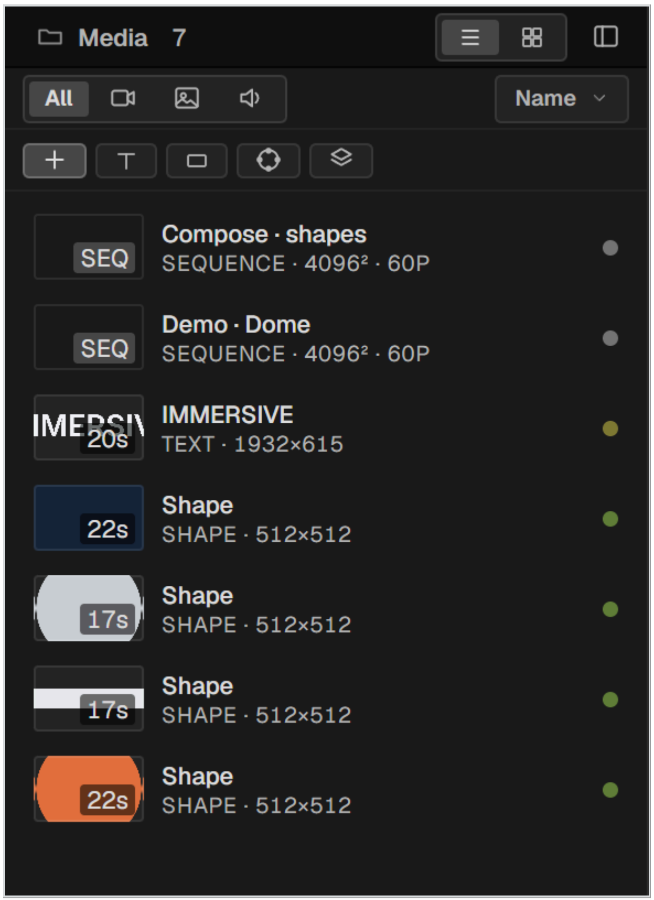
Two ways to get the picture onto a projector without the interface around it, both under `Window` and both in the `Output` menu above the viewer:

- `Full performance` — the viewer takes over the whole window.
- `Viewer-only window` — a second window you can put on another display. It shows the *complementary* view to the editor: if you are working in 2D it shows 3D, and the other way round.

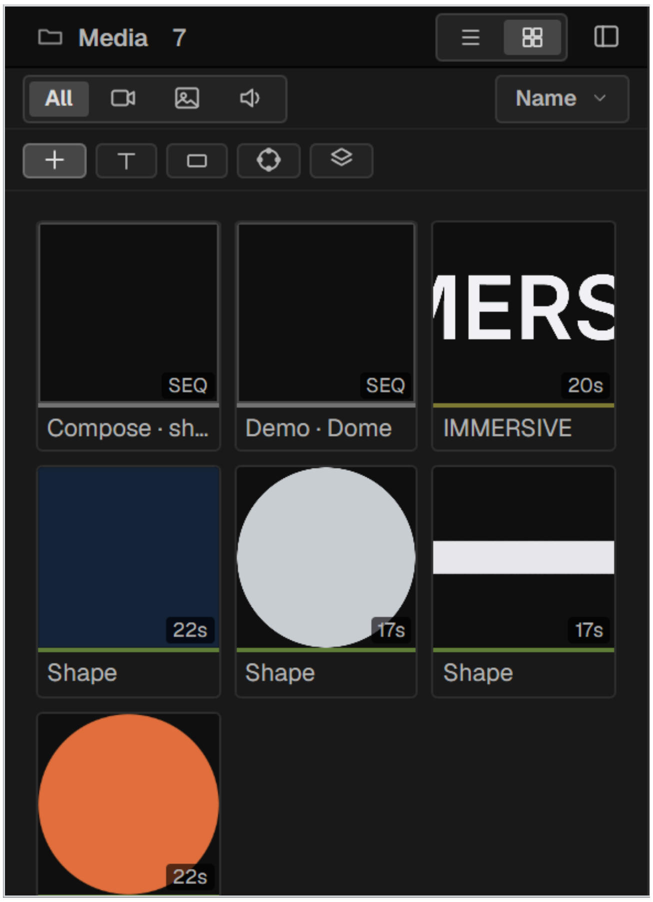
CHAPTER SIX

The media panel

The project's inventory. Everything that can become a clip lives here, including the sequences and compositions the program generates itself.



List view — name, kind, resolution and duration.



Grid view — thumbnails, for picking by eye.

The header

View

Two buttons at the top right switch between list and grid.

Filter row

All and the three kind filters — video, stills, audio.

Sort

By Name, Date or Type.

Search

Ctrl + F reveals a text filter in the filter row; Esc closes it.

The create row

Below the header: Import and four generators.

Import	Opens the file picker. Also Ctrl+I.
Text	Creates a text item — font, weight, alignment, fill, outline colour. Custom fonts can be loaded.
Shape	Creates a rectangle, ellipse or line with fill and stroke.
Compose	Builds a generated composition from the selected media. Chapter 16.
Adjust	Creates an adjustment layer — a clip with no picture of its own that applies its effects to everything below it.

Folders

Right-click an empty area of the panel to create a folder. Folders nest and can be given colours. Drag items between them. Renaming media, moving it between folders and creating or deleting folders are all undoable.

Badges on an item

A duration in brackets
: The item has in/out marks set in the source monitor, so it will enter the timeline trimmed. Chapter 10.

A coloured dot
: The item's assigned colour, which its clips inherit on the timeline.

SEQ
: The item is a sequence or a composition rather than a file.

Selecting

Click to select, Shift+click for a range, Ctrl+click to add or remove. Multi-selection matters for two things in particular: generating several proxies at once, and building a composition from a set of sources.

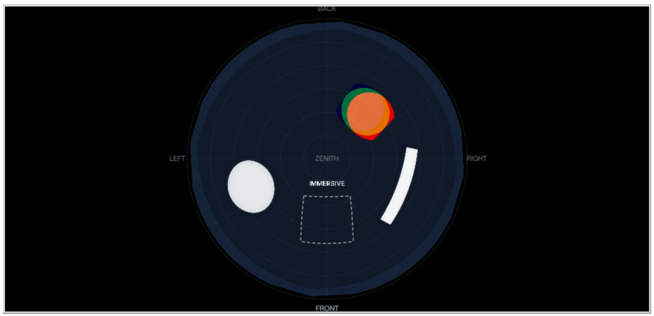
CHAPTER SEVEN

The viewer

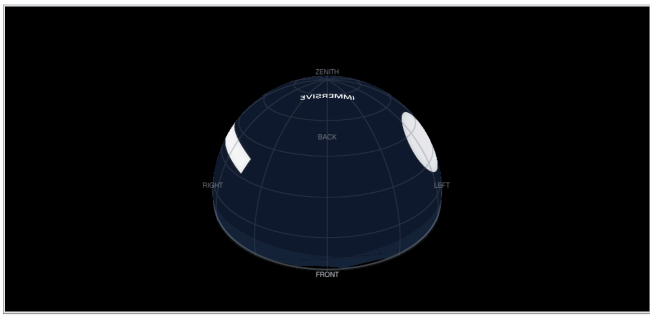
The viewer shows the composite master — the actual picture, not an approximation of it. It has two modes, and in an immersive format both are worth having open.

2D and 3D

The `2D` / `3D` switch at the top left of the viewer bar chooses between them.



2D — the dome master. The fisheye circle as it will be delivered, with orientation guides (`FRONT`, `ZENITH`) drawn over it.



3D — the preview. The same master mapped onto a dome surface, seen from inside. This is what the audience sees.

The 2D view is where you judge framing and delivery; the 3D view is where you judge whether it reads. A shape that looks reasonable as a fisheye can be badly distorted overhead, and only the 3D view will tell you.

The viewer bar

Quality
: `Full` / `½` / `¼` — preview resolution. Drop it when playback struggles; it does not affect export.

Overlays
: Grid, outline, horizon fade and, in a room, the floor rectangle.

Camera (3D only)
: `Orbit` flies around freely; `Viewer` puts the camera where a person would stand.

Zoom
: The percentage control, or the wheel over the viewer. Drag with the hand tool to pan.

`Clip` / `Comp`
: Whether the viewer shows only the selected clip or the whole composite.

`Output`
: Full performance, the viewer-only window, and the NDI and Spout broadcasts. Chapter 24.

`...`
: When the window is narrow the bar folds its less-used controls into this menu rather than truncating them.

Working directly in the viewer

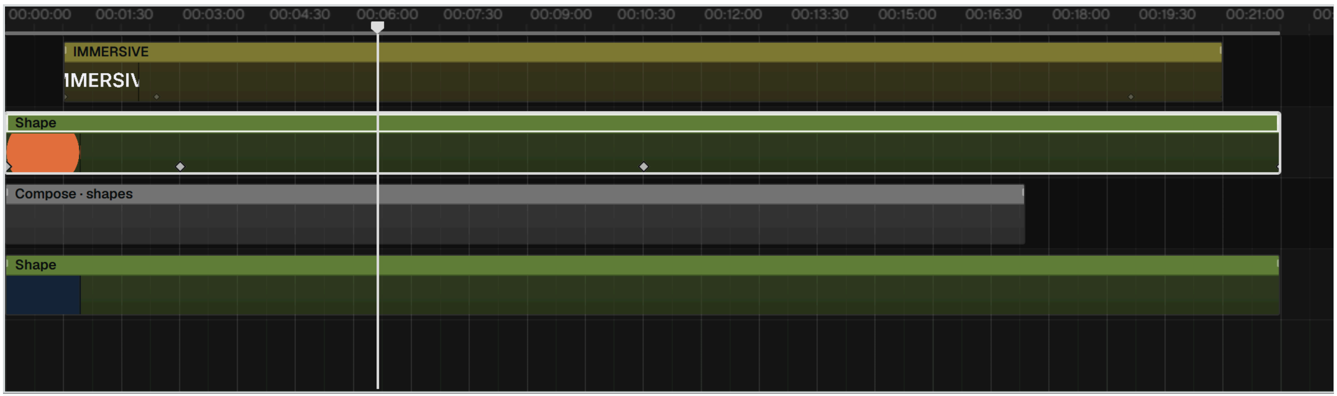
Clips can be dragged in the viewer. In a dome that moves azimuth and elevation; in a 2D or room sequence it moves x and y. Point masks are drawn and edited on the viewer canvas too — see chapter 17.

> **THE DOME CIRCLE IS THE WHOLE PICTURE**
> In a dome sequence nothing outside the inscribed circle is delivered. If part of a clip appears to spill into the corners of the 2D view, it is outside the projected image, and the export will show black — or transparency, if you exported with alpha.

CHAPTER EIGHT

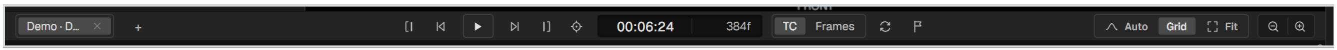
The timeline

Tracks, clips and time. The gestures are the ones you already know from a conventional editor, with two differences worth learning early: overlapping is a cut, and fades are dragged from the clip corner.



The timeline. Track headers on the left, ruler on top, clips in the body. The diamonds on a clip are keyframes.

The transport



The transport bar. Sequence tabs on the left, playback in the centre, the editing well on the right.

Play / pause	Space	Plays from the playhead. It plays whether or not there are clips there.
Go to start / end	Home · End	
Shuttle	J K L	Press J or L repeatedly for 2×, 4×, 8×. K stops.
Previous / next cut	↑ ↓	
Timecode		Click TC or Frames to switch what is displayed.
Loop		Sets the loop range from the time selection.

PLAYING OUTSIDE A LOOP

With a loop range set, pressing play *inside* it loops. Pressing play outside it plays straight through. The loop constrains playback only when you start inside it.

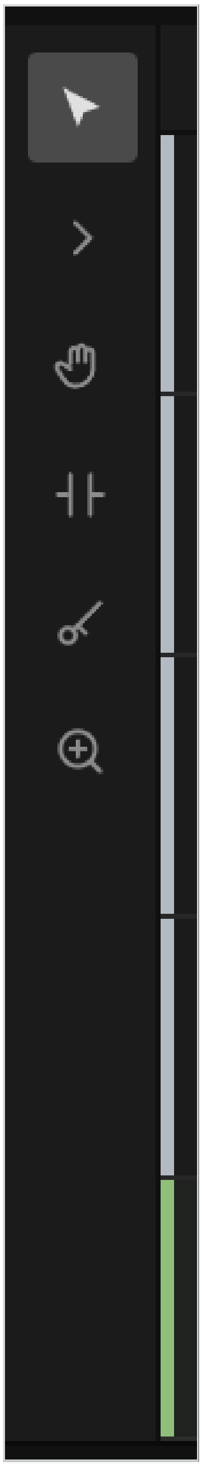
The editing well

Auto
: Automation mode — clips show their curves inline. Chapter 19.

Grid
: Snapping to the grid.

Fit
: Zooms the timeline to fit the whole project.

The tools

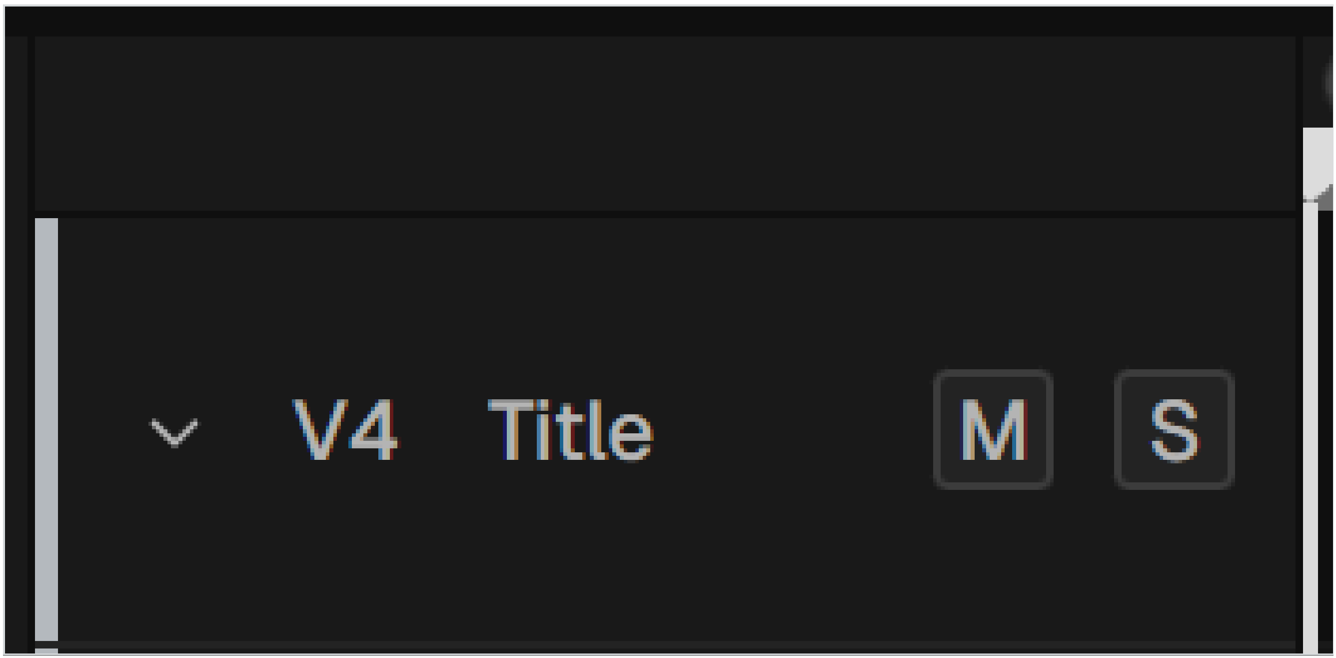


The tool rail.

Select	V	Select, move, trim.
Track select		Selects everything from the click onwards.
Hand	H	Pans the view.
Trim	T	Contextual trim: ripple, roll, slip, slide.
Razor	B	Cuts a clip.
Zoom	Z	Zooms to the click.

The status bar always spells out what the active tool does, so you never have to guess which mode you are in.

Tracks



Track headers. Name, mute and solo. Audio tracks carry a subtle tint; in a room, floor tracks are marked by a blue-framed tag.

Right-click a header to add, duplicate, rename, colour or delete a track. Drag a header up or down to reorder. Video tracks stack bottom-to-top: a clip on V3 draws over a clip on V1.

Zoom and navigation

Zoom	Ctrl+wheel, or + / −. Zooming is anchored to the pointer.
Track height	Alt+wheel.
Pan horizontally	Shift+wheel, or a horizontal wheel.
Scroll vertically	Plain wheel.

A vertical bar at the right edge of the timeline mirrors the horizontal one: its body scrolls, its end caps change the track height.

Markers

M drops a named marker at the playhead; , and . jump between them. Markers are drawn in the lower half of the ruler. There is also a command to detect beats in an audio clip and place markers on them.

CHAPTER NINE

The inspector

Everything about the selected clip, in seven collapsible sections. Which sections appear depends on what kind of clip it is — an audio clip has no Transform, an adjustment layer has no Source.

Transform	Where the clip sits. Azimuth, elevation, size and rotation in a dome; x, y, scale and rotation in 2D and rooms. Plus `Mirror`.
Clip	Opacity, blur, crop, mask, blend mode, black key. Chapter 17.
Source	How the footage is interpreted: `Fulldome src`, `Equirect 360°`, `Fisheye`. Chapter 3.
Playback	Speed and looping. Chapter 11.
Color	Exposure through to LUT. Chapter 18.
Motion	Procedural movement. Chapter 19.
Effects	The effects chain for this clip. Chapter 20.

Parameter rows

Every numeric parameter is one row: a name, a coloured track, a value box and a keyframe diamond.

- Drag the track to change the value.
- Double-click the value box to type an exact number.
- Click the diamond to set a keyframe at the playhead; click a filled diamond to remove it.
- The track's colour identifies the parameter, and the same colour is used for its automation curve.

> **EDITING A VALUE WRITES A KEYFRAME**
> If a parameter is already animated, changing its value creates a keyframe at the playhead rather than shifting the whole curve — the same rule as After Effects. If it is not animated, the value simply changes.

The header

The top of the inspector shows the clip's thumbnail, name and kind, and a `Source` button that opens the source monitor on that media.

The Reactive FX tab

The second tab holds the audio-reactive engine and the effects chain. It is documented in chapter 20.

PART THREE

Editing

Getting material in, cutting it, and placing it in each of the three formats.

CHAPTER TEN

Importing and managing media

Three ways in

Import media… — the button, Ctrl + I, or a double-click on empty panel space

Every image becomes its own clip. It never asks about sequences.

Import image sequence…

Groups numbered files into a single sequence item and asks for the frame rate. There is a third answer to that question — As separate images — for when the numbering is a coincidence.

Dragging files onto the panel

Never groups. Each file is its own item.

> **THE CHOICE IS MADE BEFORE THE PICKER OPENS**
> Whether numbered images become a sequence is decided by which command you use, not by a checkbox afterwards. This is the Premiere model.

Proxies

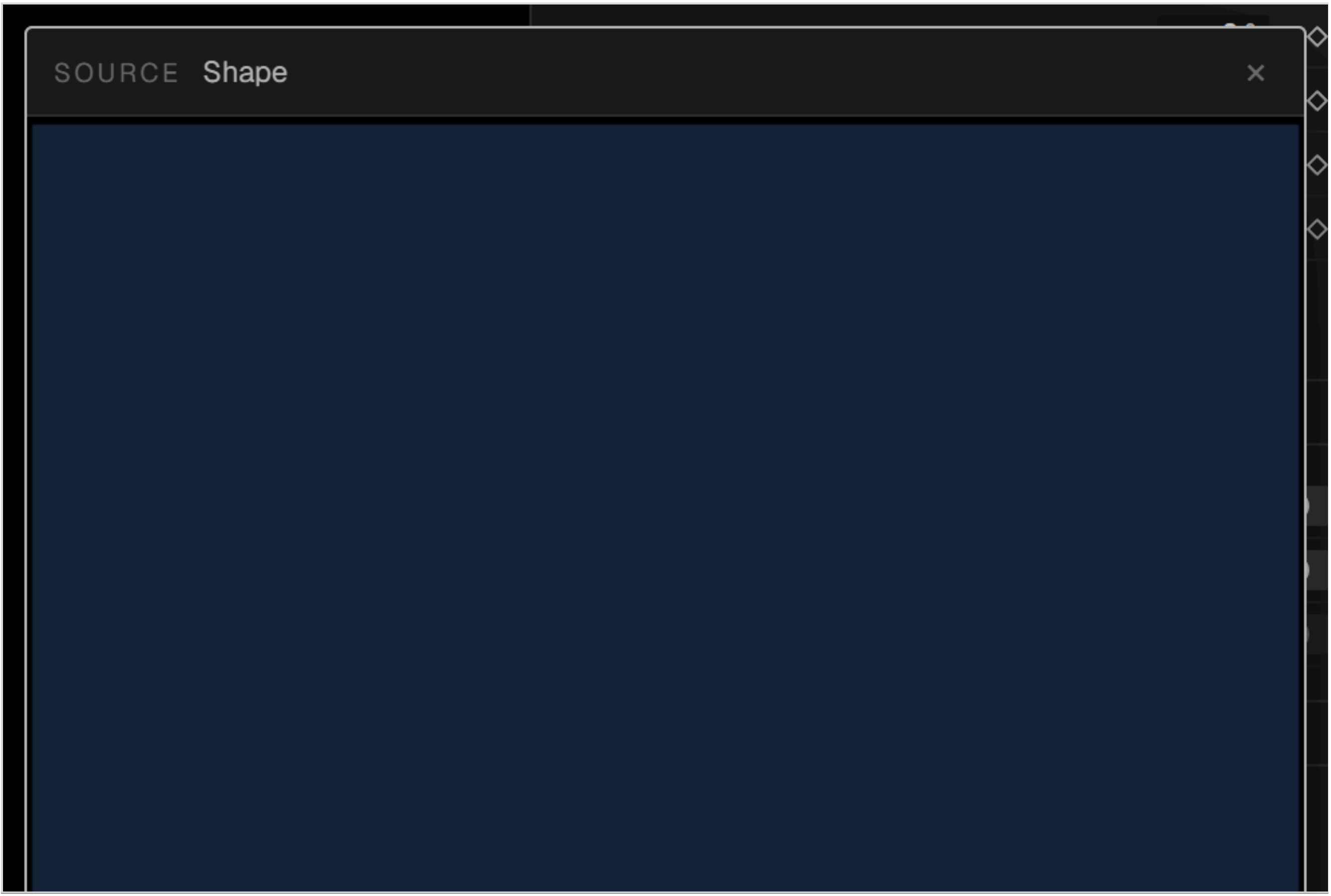
A proxy is a light, all-intra copy of a heavy clip used for editing. In this program proxies are **manual**: nothing is transcoded behind your back.

- Right-click a media item and choose Generate proxy. Select several first to queue a batch.
- When you import something heavy, the status bar reminds you the option exists. It does not act.
- Existing proxies on disk are adopted automatically, both when a project is reopened and when the same footage is imported again.
- A proxy whose generation was interrupted is detected and discarded rather than used.

> **WITH MASTERING FOOTAGE A PROXY IS NOT AN OPTIMISATION**
> Long-GOP camera masters can take over a second per frame to scrub. A proxy brings that to a few milliseconds. For that kind of material it is not a nicety — it is the difference between editing and waiting.

The source monitor

Double-click a media item — or press Source in the inspector — to open the source monitor: a floating player for reviewing material and marking the part you want.



The source monitor. Transport, scrub bar, in and out marks and an insert button. The picture itself is the drag handle: what lands on the track is only the marked range.

- I and O set the in and out points, Space plays, the arrow keys step a frame at a time.
- The window is deliberately not modal — you can keep working with it open. Move it, resize it from the corner, close it with Esc.
- Marks belong to the *media*, are saved in the project, and are respected when you drag from the media panel too. An item that will enter trimmed shows its duration in brackets.
- Simply looking does not change anything: marks are only committed by an explicit gesture.

Relinking and replacing

Relink
: If a file has moved, point the item at its new location; every clip using it follows.

Replace media
: Swaps the footage under existing clips, keeping their position, duration, effects and automation. If a clip was looping, it loops over the *full duration of the new media* while keeping its length on the timeline. This applies inside nests and compositions too.

Text and shapes

Text items carry font, weight, alignment, fill and outline colour, and can load custom font files. They are rendered at a generous internal resolution so they stay sharp on a 4K dome; there is no pixel-size field because the proportion of the canvas is what matters, not the point size.

Shape items are a rectangle, ellipse or line with a fill and a stroke.

Saving

Projects are `.isp` files — JSON, written atomically with a `.bak` alongside. The program also autosaves, keeps a recovery history, and can save incremental versions (`_v01`, `_v02`...) from the `Save` button's right-click menu.

CHAPTER ELEVEN

Editing in the timeline

Moving clips

Drag a clip anywhere on its body to move it. A ghost shows where it will land. Alt+drag copies instead of moving. Linked audio and video move together, and a linked partner is constrained to horizontal movement so the pair cannot be separated onto mismatched tracks.

Overlapping is a cut

When you drop a clip on top of another, the moved clip **trims the stationary one** — and splits it in two if it lands in the middle. Nothing is destroyed: the trimmed material is still there, and dragging the clip away again does not restore it, but the source is untouched. No crossfade is created automatically.

Trimming

Handles
With the select tool, drag either end of a clip.

Contextual trim (T)
The trim tool picks the operation from where you grab: **ripple** at an outer edge, **roll** on a cut, **slip** in the middle of a clip, **slide** below it. A linked partner is trimmed to match.

Fades and crossfades

Every clip has a small handle in its upper corners.

- Drag it **inwards** for a fade in or out.
- Drag it **outwards over a cut** for a manual crossfade with the neighbouring clip. Video crossfades geometrically; audio crossfades by gain.

Cutting and deleting

Razor	B	Click to cut.
Split at playhead	Ctrl+E	Cuts the selection, or every clip under the playhead.
Delete	Del	Leaves a gap.
Ripple delete	Shift+Del	Closes the gap.

Copy and paste

Copying works on the whole selection, not one clip. When you paste, each clip lands on its own track at its original spacing relative to the first — a multi-layer arrangement pastes back intact. Linked pairs are relinked only if both halves were copied.

Copy and paste attributes

Ctrl+Alt+C and Ctrl+Alt+V carry everything that belongs to a clip — transform, looping, speed, colour, masks, effects, motion and all automation — from one clip to another. It works in both directions between an ordinary clip and a composition, because a composition *is* a clip.

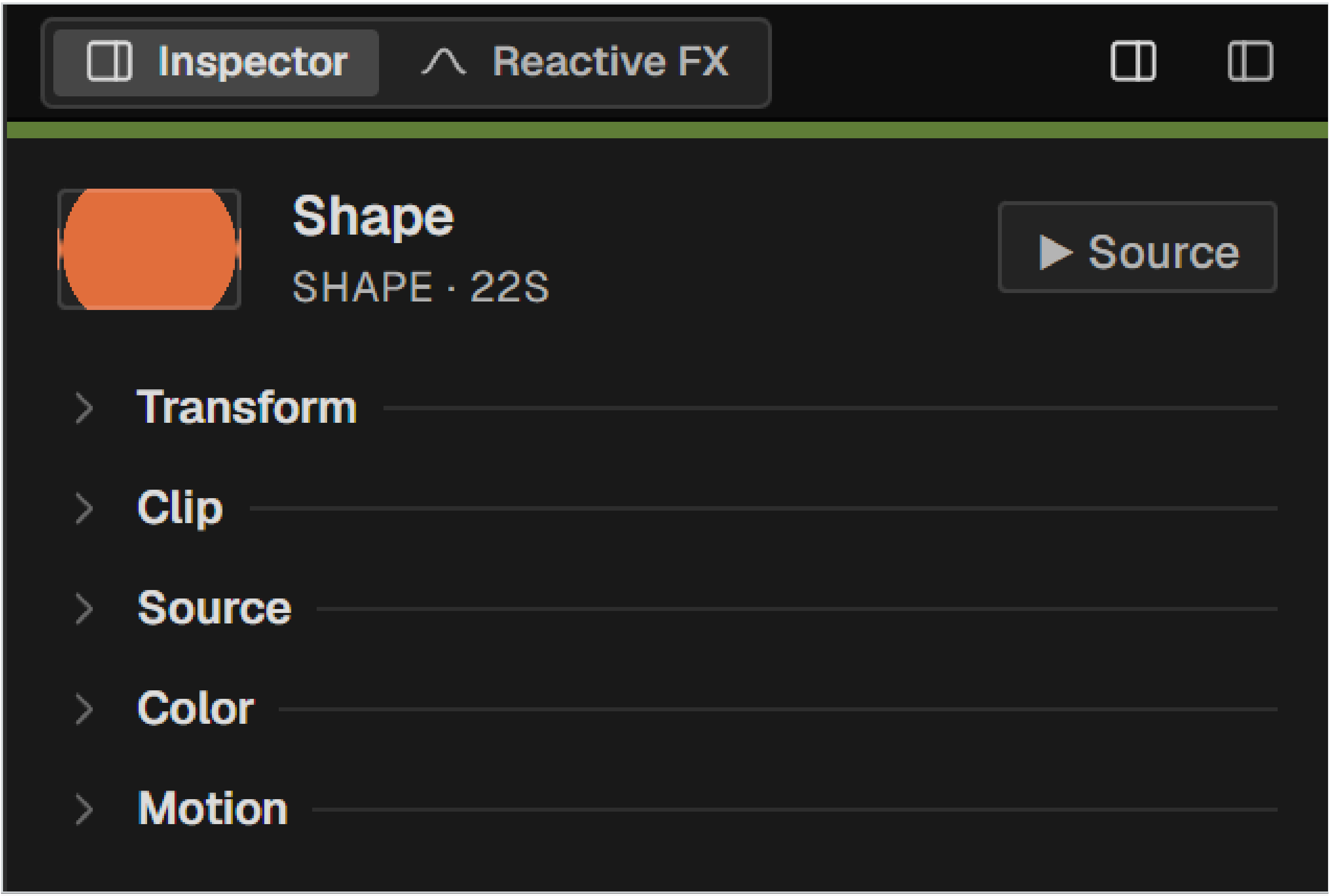
What does not travel: identity, position, duration and the media itself. **The destination always keeps its place and its length on the timeline.** Neither does a composition's generating configuration.

> **AUTOMATION IS TRIMMED TO FIT**
> Paste from a 40-second clip onto a 15-second one and only the automation that fits is kept. If the first surviving keyframe is not at the start, the interpolated value is inserted there, so the curve does not begin with a jump.

Selection, snapping and grouping

- Drag on empty timeline space for a marquee; drag on the ruler for a time selection, which becomes the loop range.
- Snapping catches clip edges, the playhead, markers and the grid. The Grid button in the editing well toggles the grid component.
- Ctrl+D duplicates, Alt+←/→ nudges by a frame.

Speed and looping



The Playback section. Speed, and the looping controls.

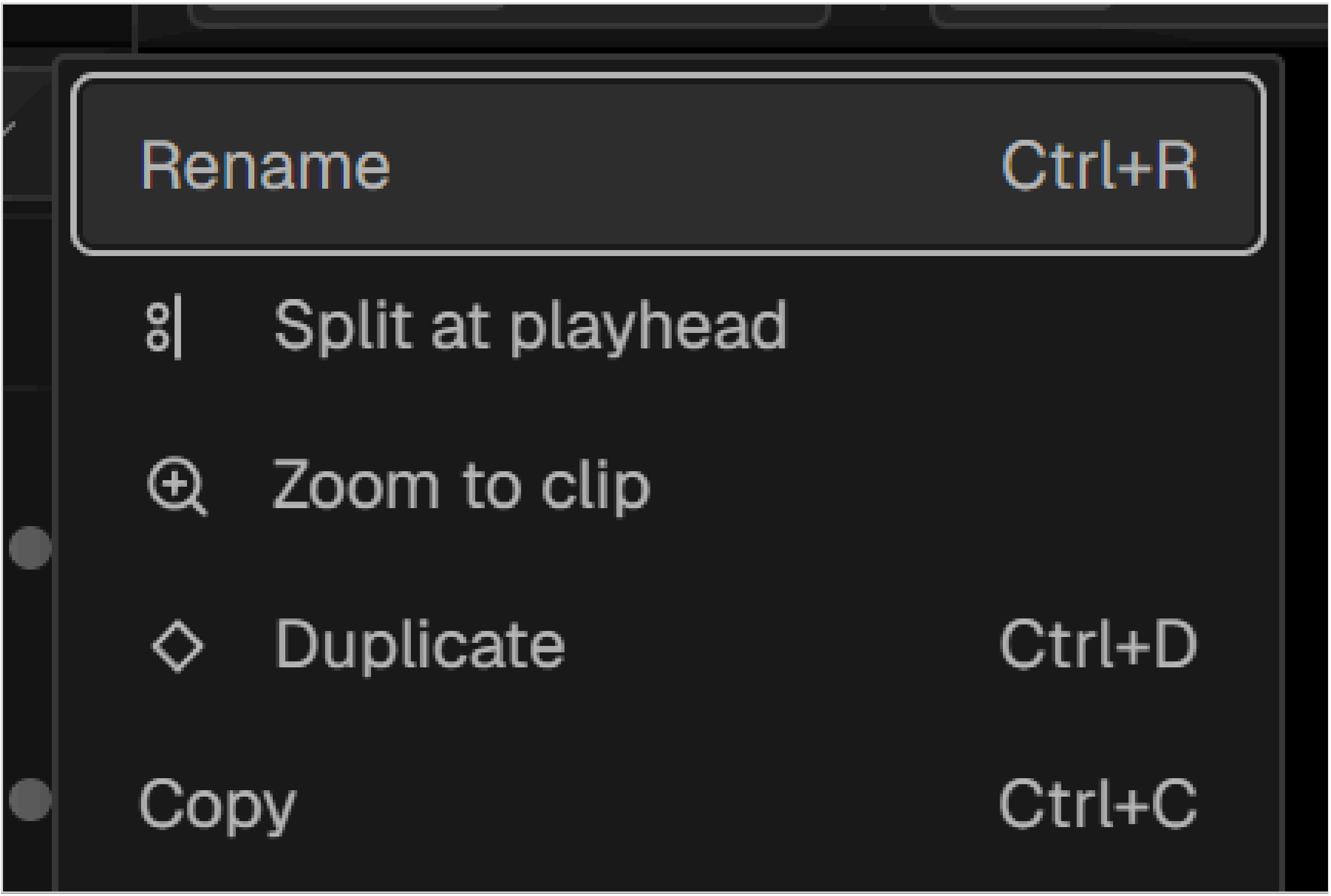
Speed
: 50–200 %. Changing it stretches the clip *and its automation* together, so a movement you have keyed stays in the same place relative to the picture.

Loop
: Repeats the source for the length of the clip. The inspector shows the cycle length and which part of the source it repeats, in timecode.

`From clip`
: Re-reads the cycle length from the clip's current duration.

> **DO NOT SWITCH LOOPING OFF TO CHANGE THE CYCLE**
>
> Turning the loop off trims the clip back to whatever is left of the file. Use `From clip` to redefine the cycle instead — it changes the cycle length and nothing else.

The clip context menu



Right-clicking a clip. The per-clip operations, including nesting, render in place and attribute copying.

CHAPTER TWELVE

Placing clips in the dome

In a dome sequence a clip is placed by direction. This chapter covers the four transform parameters, the three source switches, and how to think about a curved surface.

The transform parameters

Azimuth	Compass bearing around the horizon, in degrees. 0° is the front of the dome.
Elevation	Height above the horizon, in degrees. 90° is the zenith, directly overhead. Negative values go below the horizon, which is real projected area in a dome with coverage above 180°.
Size	How much of the dome the clip subtends.
Rotation	Roll about the clip's own centre.
Mirror	Flips the clip horizontally.

All four can be dragged in the row, typed exactly, keyframed and animated. Dragging the clip in the viewer moves azimuth and elevation together.

What a curved surface does to a picture

A rectangle placed on a dome is not a rectangle any more, and the distortion grows with size and with height. Two consequences to keep in mind:

- A large clip near the zenith stretches noticeably. Judge it in the 3D preview, not the fisheye.
- Near the zenith, a degree of azimuth covers very little distance. This is why the `Float` motion preset animates in the dome plane rather than in azimuth and elevation: otherwise the same drift would look different depending on where the clip sat.

The source switches

Chapter 3 explains what these mean; this is how to use them.

`Fulldome src`
: Turn on when the footage *is* a dome master — a circular fisheye filling the frame. It is drawn one-to-one; position and size no longer apply in the usual way.

`Equirect 360°`
: Turn on for equirectangular 360° footage. It is reprojected into the dome.

`Fisheye` + `Amount`

Bends a fulldome source further. Only available when `Fulldome src` is on — the row is greyed out otherwise, with a tooltip explaining why. Turning `Fulldome src` off also turns this off.

> **COMPOSITIONS ARE FULLDOME SOURCES**
> A generated composition covers the whole dome by nature, so its clip is created with `Fulldome src` already on. You do not place it; you place the things inside it.

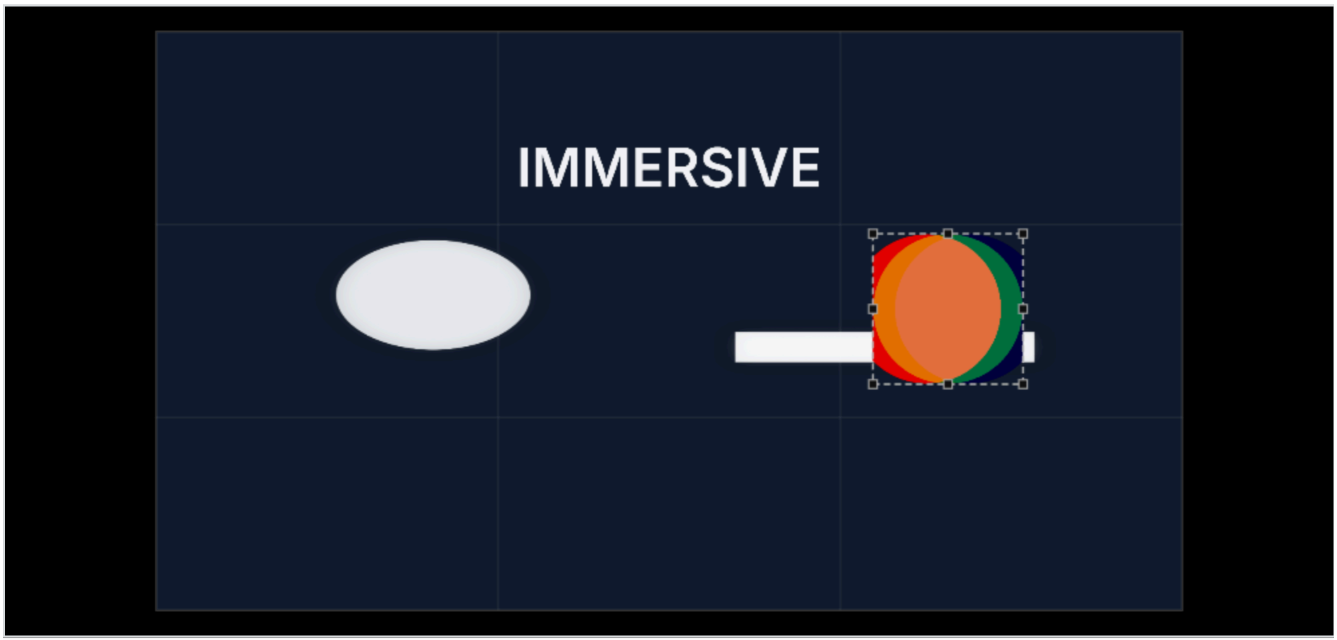
Adjustment layers

An adjustment layer is a clip with no picture. It applies its effects chain to everything composited below it, across the full dome. It has no Transform and no Source section — by contract it is always a full-frame fulldome clip — but it carries the complete effects catalogue.

CHAPTER THIRTEEN

2D sequences

A 2D sequence is an ordinary rectangular canvas. Everything in the program works the same way; only the placement model changes.



A 2D sequence. The canvas is a rectangle, with the safe grid drawn over it.

What changes

Transform	`X`, `Y`, `Scale` and `Rotation` instead of azimuth, elevation and size.
Source section	The dome switches do not apply.
Viewer	The 3D preview is hidden — there is nothing to preview in 3D.
Motion presets	A flat set: `Rotate`, `Pulse`, `Horizontal`, `Vertical`, `Float`.
Compose	The flat layouts — grid, line, row, column, weave — rather than the dome ones.

Why you will use 2D even for dome work

Two reasons, both structural:

- **Compositions are built flat.** The `Weave` composition, for instance, lives in a 1:1 flat nest — that is where neighbouring strips meet at exact right angles — and enters the dome as a fulldome source with fisheye applied to the whole assembly at once.
- **The floor of a 360° room** is edited as a flat area.

You can add a 2D sequence to a dome project at any time from `Project → New sequence…`; formats mix freely within one project.

CHAPTER FOURTEEN

The 360° room

A room is several projection surfaces treated as one piece: walls unrolled into a single wide strip, an optional floor, a 3D preview from the audience position, and an export that can deliver the whole thing or one projector's worth at a time.

Setting up the geometry

The room is defined when the project is created, and can be reconfigured at any time from `Project → Room geometry…` .

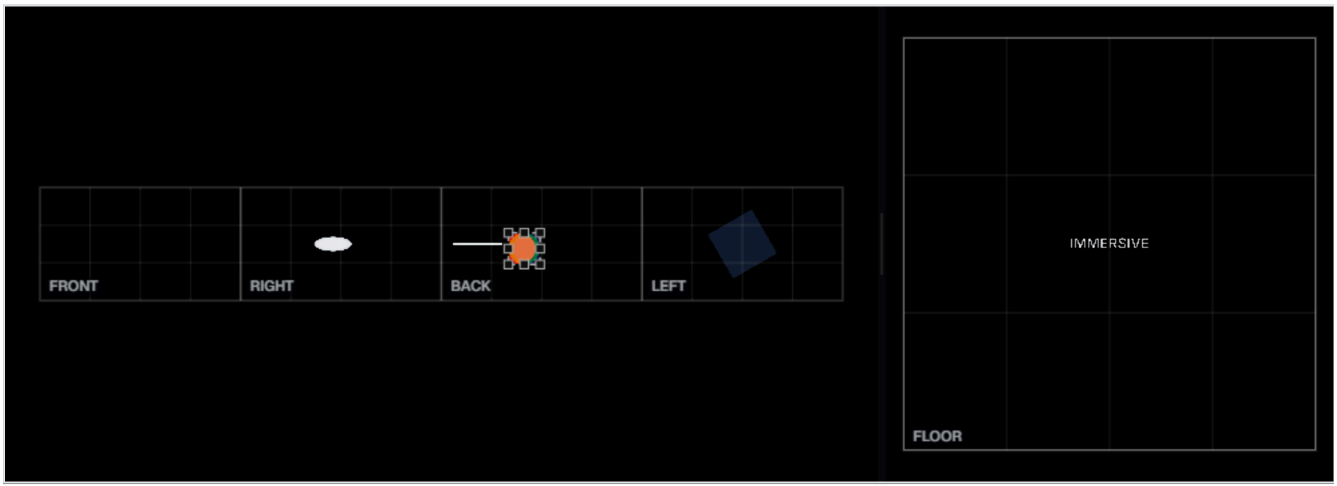
Room setup. Walls with their roles, physical measurements and pixel resolutions, the floor, and a plan view that shows whether the walls actually close into a room.

Each wall has a role (front, right, back, left), a physical width and height in centimetres, and a pixel resolution. The plan preview solves the wall loop into a floor plan and will tell you if the geometry does not close, or if it closes by crossing itself.

CENTIMETRES ARE FOR THE 3D VIEW; PIXELS ARE THE CANVAS

The strip you edit is measured in **pixels** — its width is the sum of the walls' pixel widths. The physical measurements exist so the 3D preview has the right proportions and so the floor can be derived correctly. Getting the centimetres wrong makes the preview lie; getting the pixels wrong makes the delivery wrong.

Editing the strip

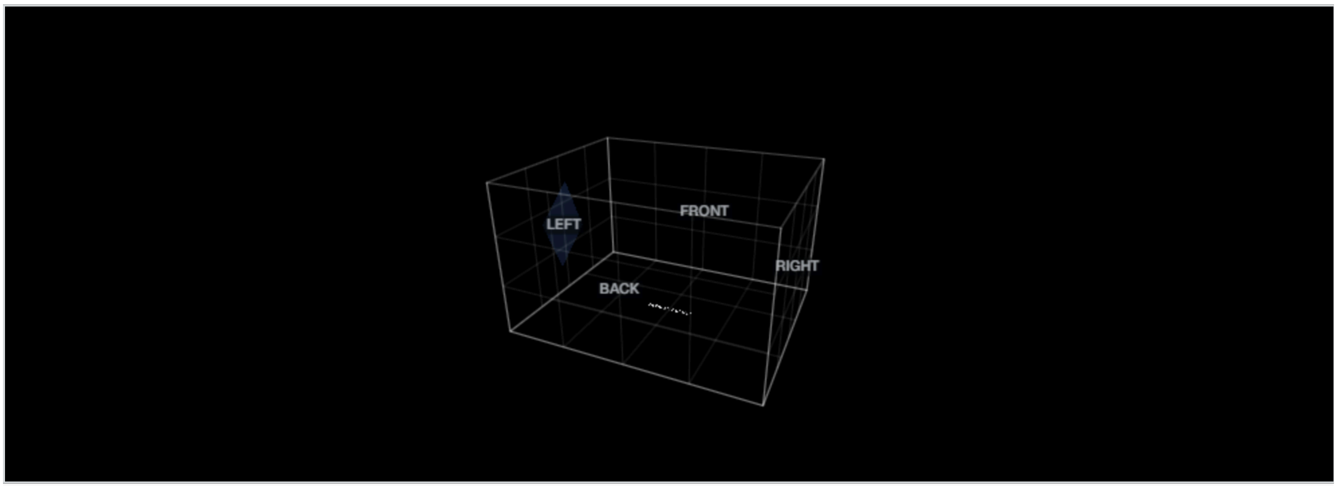


The unrolled strip. All the walls side by side as one canvas, with seams and wall labels drawn over it. The floor is the band along the bottom of the same canvas.

You edit this exactly like a 2D sequence: clips are placed by x, y and scale, and a clip can span a seam and run across two walls. The seams and labels are guides, not boundaries.

Floor content goes on floor tracks, identified by a blue-framed tag in the track header. The floor is part of the same canvas rather than a separate sequence, so one composite covers everything.

The 3D preview



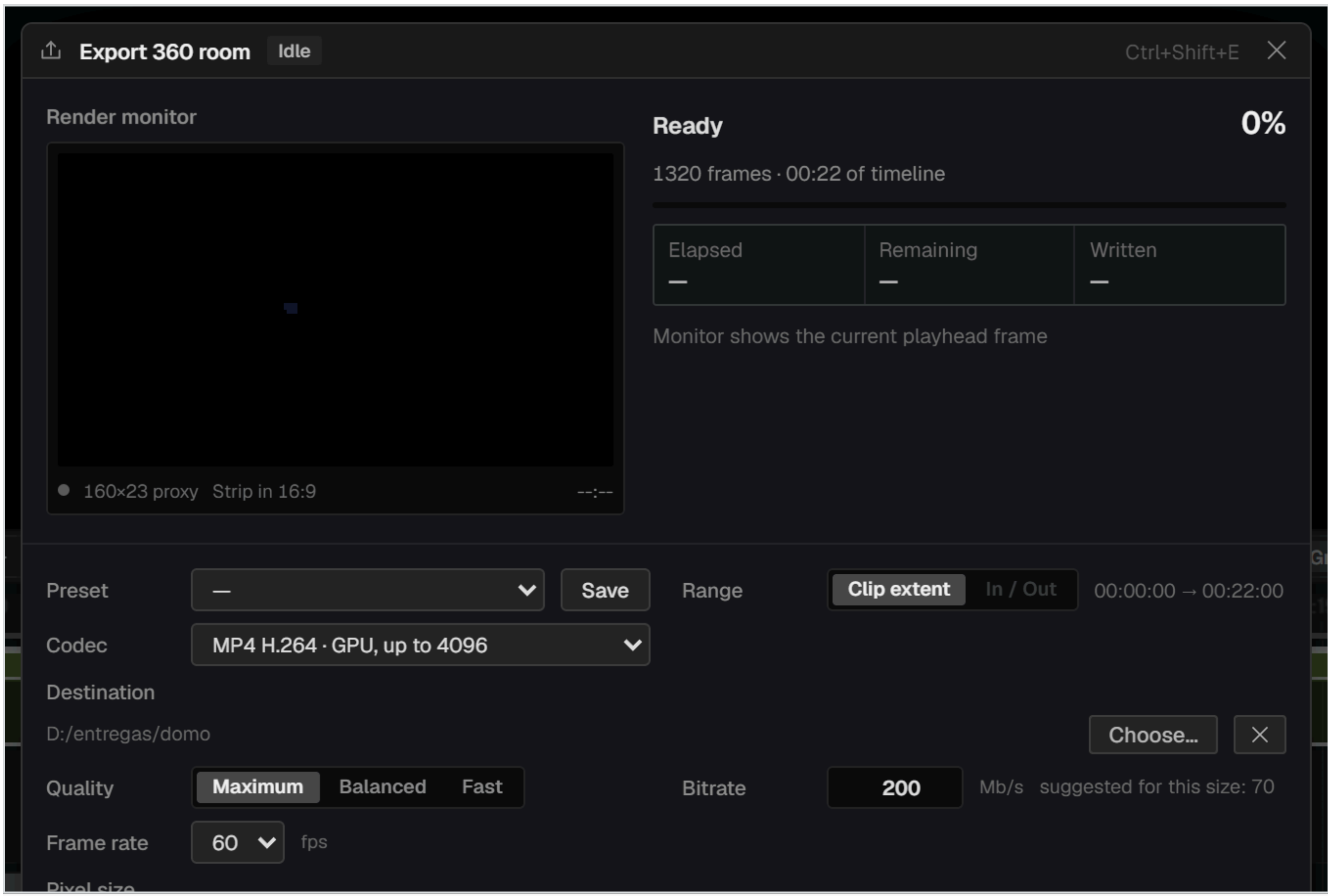
The room in 3D. Walls and floor as they will be seen, with the camera either orbiting freely or standing where a person would.

The content is drawn unshaded on purpose: this view is a preview of what will be projected, and a decorative light would misrepresent the brightness. The shading you see belongs to the translucent shell drawn around the outside.

Exporting a room

The export sheet offers three modes for a room:

Full strip	The walls only, as one wide file.
Strip + floor	Two files: the wall strip and the floor.
Each wall + floor	One file per wall plus the floor — one per projector. Only offered when the room has a floor.

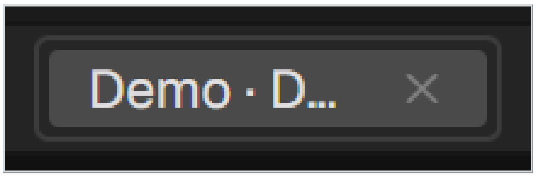


Exporting a room. The mode selector adds the room-specific jobs to the queue.

Sequences and nests

A sequence is a media item. That is the whole mechanism, and it gives you tabs, nesting and compositions at once.

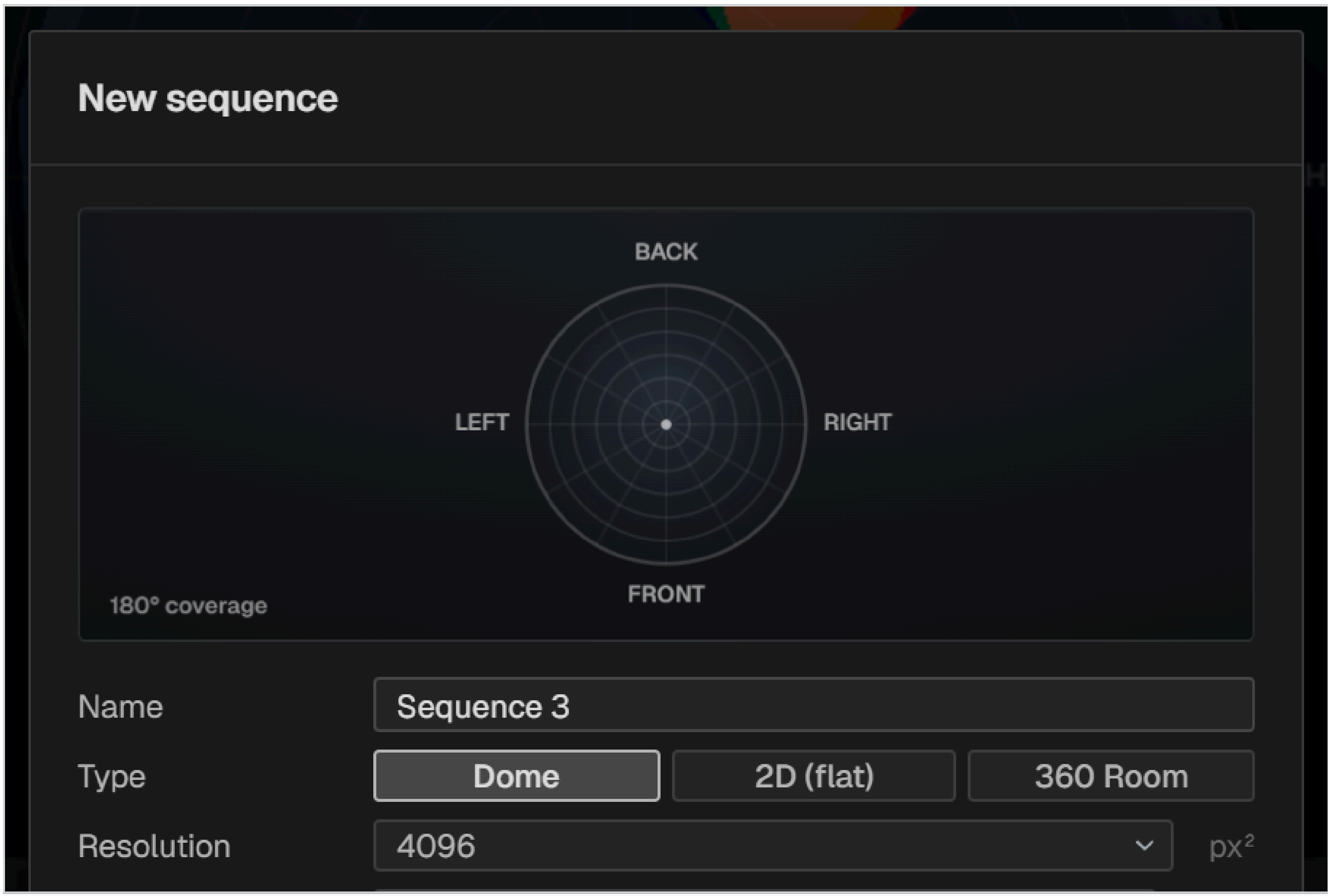
Sequence tabs



Sequence tabs sit at the left of the transport bar.

Tabs are a fixed width so the bar does not shift as you rename things; three are visible at a time and the rest are reached with the wheel over the tabs. Drag a tab to reorder. + creates a new sequence.

Creating a sequence



New sequence. Dome, 2D or 360 Room, each with its own parameters and a preview. The new sequence joins the project without disturbing anything already in it.

Nesting

Select some clips and choose `Nest selection` from the `Edit` menu or the clip's context menu. They are replaced by a single clip that contains them. Double-click it to go inside; the tab bar shows where you are.

- A nested clip is an independent copy — editing it does not affect the originals.
- In a dome sequence, a nest is created as a fulldome source: it composites internally and enters the dome as a complete dome image.
- If the content of a nest is shortened, its instances in the parent sequence shorten with it rather than holding a frozen last frame.

Audio inside a nest

The rule is that no sound plays unless it is visible on an audio track. A nest containing audio tracks therefore creates a *derived* audio clip in the parent, linked to the video clip. It moves, trims and unlinks like any A/V pair, and you can see exactly what is making sound.

Composition proxies

A nest can be cached as a single video, so playback decodes one stream instead of many. On a heavy composition this is the difference between a slideshow and something you can work with. The cache invalidates itself when the composition's contents change.

Render in place

Right-click a clip or a nest and choose `Render in place...` to bake it to a silent MP4 and replace the timeline instance with the baked file on a new track. Useful for freezing an expensive composition once it is final. A progress view shows the live frame, a bar and an estimate, and can be cancelled.

PART FOUR

Creative tools

The generators, the image treatments and the two ways of making things move.

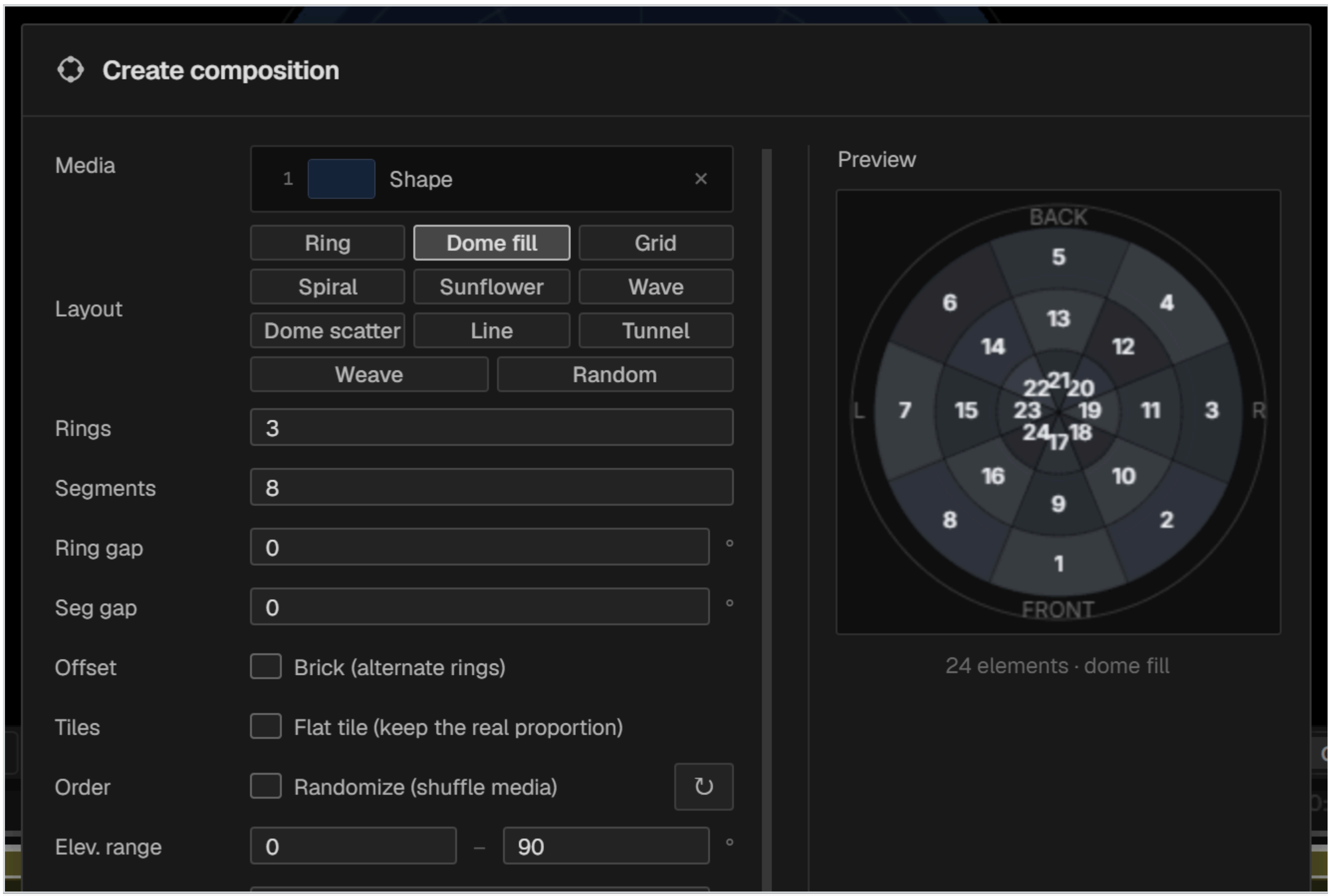
CHAPTER SIXTEEN

Compose

Compose builds an arrangement out of your own footage — a ring of clips around the horizon, a tunnel rushing towards the viewer, a woven basket filling the dome — and hands it back as an editable nested sequence rather than as an effect you cannot get inside.

Making one

1. Select the media you want to use in the media panel.
2. Click `Compose` in the create row.
3. Choose a type and set its parameters. The dome preview updates as you work.
4. Confirm. The composition lands on the nearest track with room for it.



The Compose dialog. Type across the top, the media basket, then the parameters for the chosen type. The preview is live, and the editor stays visible behind the dialog so you can judge the composition against what is already on the dome.

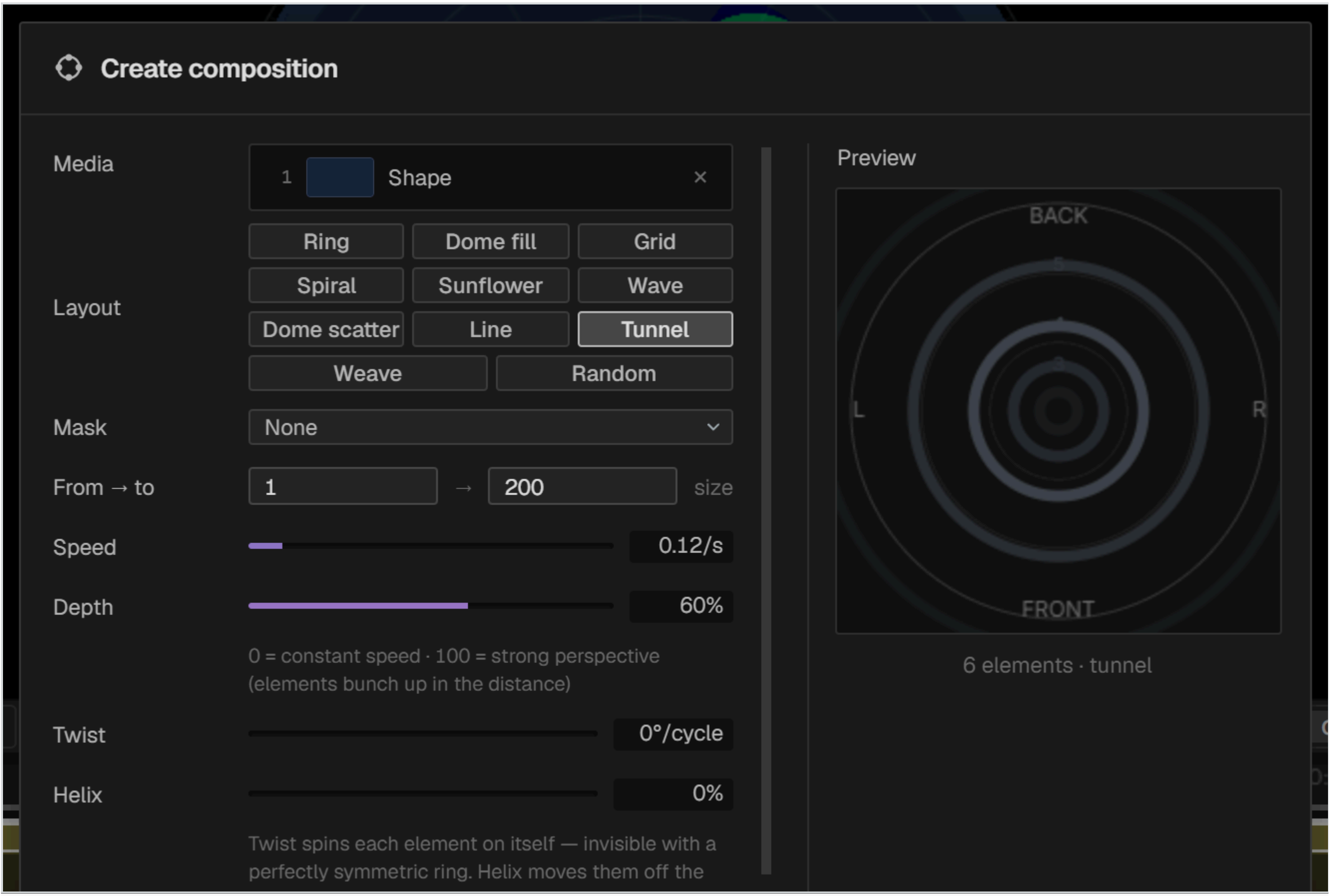
The media basket

The `Media` field shows what the composition *contains* — thumbnails, names and order — not a checklist of the whole project. Remove with `×`; add by dragging from the media panel or by double-clicking there. Order matters: it decides which source goes where.

The types

TYPE	WHAT IT MAKES
Ring	A circle of clips at one elevation.
Dome fill	Fills the dome with a grid of rings and segments. Tiles are not distorted.
Grid	A rectangular grid.
Spiral	A spiral from the zenith outwards.
Sunflower	Phyllotactic packing — even coverage with no visible rows.
Wave	A wave-shaped band.
Dome scatter	Even quasi-random scattering over the hemisphere.
Line	A straight line of clips.
Tunnel	Depth: elements grow from the centre and pass the viewer.
Weave	Interlaced strips travelling endlessly, like basketry.
Random	Random placement with a jitter control.

Tunnel



Tunnel. Sources are 1:1 images with alpha, marked as fulldome, that start small and grow until they leave past the edge of the dome.

No particular shape is assumed — a ring reads as a legible tunnel, but any distribution of alpha works. Each element is phase-shifted so the stream is continuous. The controls are:

`From` / `To`
: The size range each element travels through.

`Speed`
: How fast.

`Depth`
: Bends the growth curve. An exponential rise is what reads as perspective — approaching a viewer at constant speed multiplies the radius rather than adding to it.

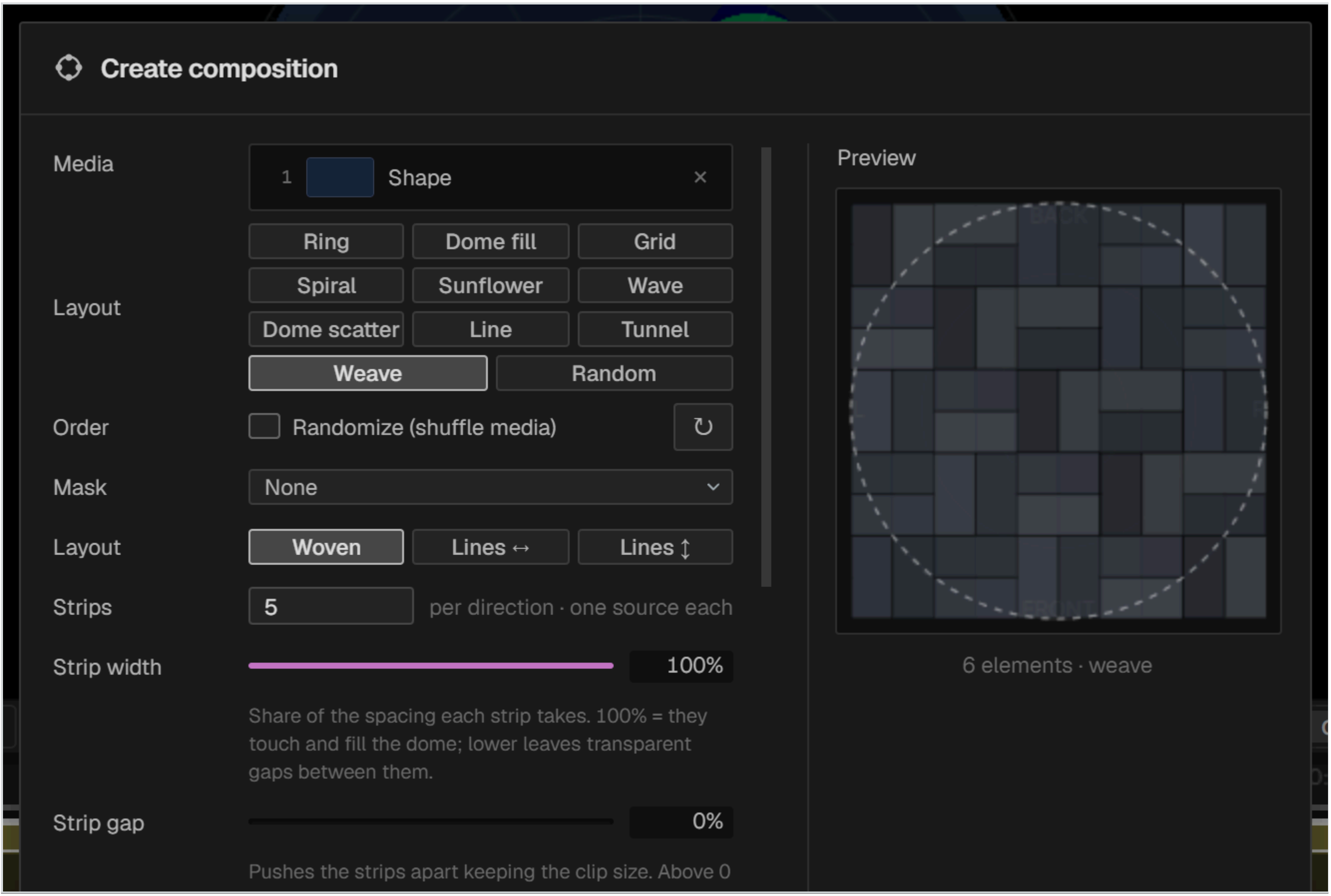
`Spin`
: Degrees of rotation per cycle. Invisible with a perfectly symmetrical source.

`Helix`
: Pushes elements off the axis in a spiral. Visible with any source.

`Fade in` / `Fade out`
: Separate envelopes, each with its own amount as a percentage of the cycle.

Elements are drawn in depth order every frame, since their ranking by nearness rotates as they cycle.

Weave



Weave. Two families of crossing strips travelling endlessly. Built in a flat 1:1 nest and brought into the dome as a single fulldome source.

Weave never distorts its sources: the side that crosses the strip takes the strip's thickness and the other follows the media's own proportions. One source per strip. The interlacing — which strip passes over which — is decided per crossing rather than per clip, so the pattern stays still while the strips flow through it.

Layout
: Woven, Lines ↔ or Lines ↕.

Strips
: 1–24 per direction, one source each.

Strip width
: Percentage of the step. 100 % means strips touch and fill the dome; less leaves a transparent gap. Width and spacing are the same control.

Packing
: Capped at 100 so one clip begins where the last ends and clips are never cut.

Direction
: One way, The other, Alternate or Still.

Order
: Shuffles which source goes to which strip.

The composition mask

Every composition can crop its elements to a shape, with a size from 0 to 100 %. It is the quickest way to turn square footage into a ring of circles.

Editing a composition afterwards

Two surfaces, and they agree with each other:

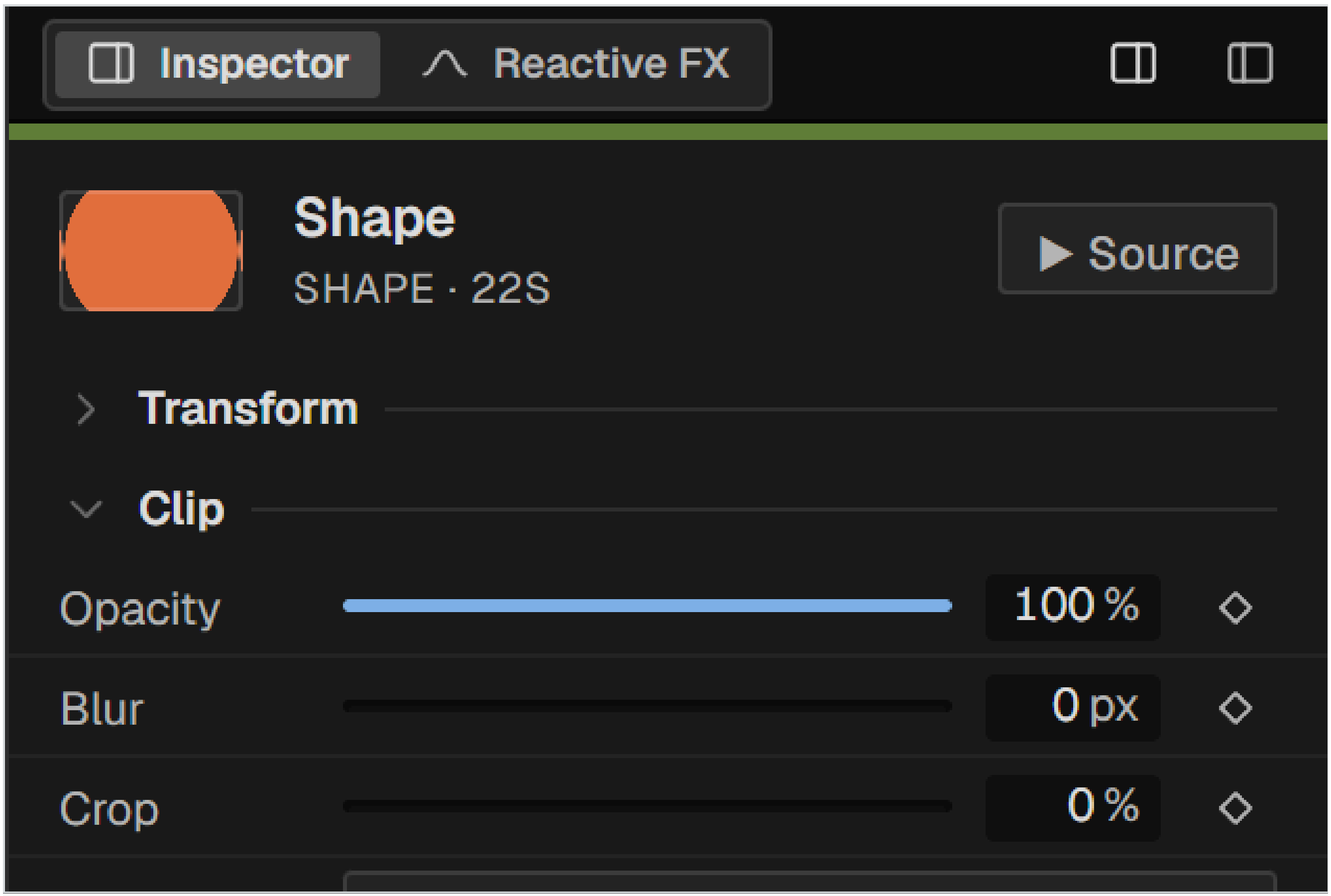
- **The inspector** shows the type, the two most useful parameters for that type, and `More options…`.
- **The dialog** reopens the full parameter set.

And because the composition is a real nest, you can also go inside and move individual clips by hand. Manual changes are preserved when the composition is regenerated.

> **WHERE A COMPOSITION LANDS**
> It goes to the nearest track with a free gap for its duration, measured from the selected track header — or from the bottom if none is chosen. Only if there is no room does it create a new track.

Effects and masks

The Clip section



The Clip section holds the per-clip image treatment.

Opacity	0–100 %.
Blur	In pixels.
Crop	Trims the edges of the source.
Mask	A shape or an image mask — see below.
Point mask	`Add mask` starts a drawn mask.
Blend	Normal, Add, Screen, Multiply, Darken, Lighten.
Remove black	Keys out a black background — for footage delivered without alpha.

Masks

Shape masks

The `Mask` dropdown offers `Circle`, `Rounded`, `Diamond` and `Vignette`, plus `Custom` for a PNG used as a matte. A size control scales the mask, and a feather softens its edge.

Point masks

`Add mask` lets you draw a mask by clicking points, **directly on the viewer canvas** — the same place you see the picture. A clip can carry several, each with its own invert and feather. This is the tool for isolating part of a shot rather than cutting it to a geometric shape.

The effects chain

Effects live in the `Effects` section of the inspector and in the `Reactive FX` tab — the same chain seen from two places. Add with `Add Effect`, reorder by dragging, switch one off with its power button, remove it with the bin.

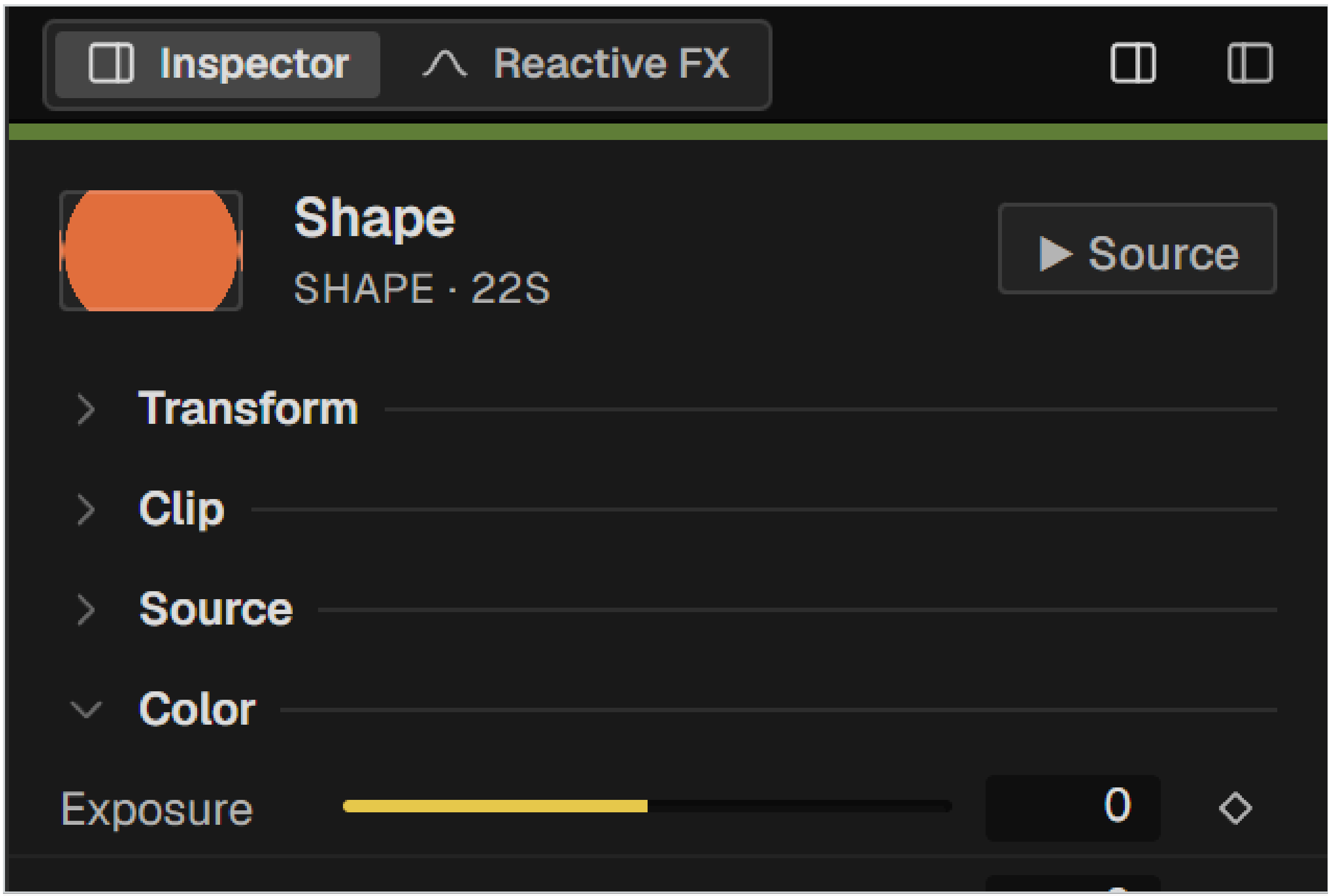
Every effect parameter can be keyframed, and every effect can additionally be driven by audio. Chapter 20 covers the audio side; chapter 28 lists all 26 effects with their parameters.

> **EFFECTS ON EVERYTHING BELOW**
>
> To treat the whole picture rather than one clip, use an adjustment layer: an effects chain that applies to everything composited under it. Create one from `Adjust` in the media panel's create row, or from `Add Adjustment Layer` in the Reactive FX tab.

CHAPTER EIGHTEEN

Colour

Colour is per clip and runs entirely in the shader, which means it costs nothing extra and applies identically in the viewer, the live outputs and the export.



The Colour section: primary sliders, three-way wheels, tone curves and a LUT slot.

The sliders

Exposure	Overall brightness.
Contrast	
Saturation	
Temp	Warm / cool.
Tint	Green / magenta.
Glow	A soft bloom around highlights.
Chroma	Chromatic aberration.

All seven can be keyframed and animated.

Lift, Gamma, Gain

The three wheels are a primary grade in the usual arrangement: `Lift` for shadows, `Gamma` for midtones, `Gain` for highlights. Drag inside a wheel to push the colour, use the bar beneath it for level.

Curves

A tone curve with `Luma`, `R`, `G` and `B` channels. Click to add a point, drag to shape, `Reset` to straighten.

LUTs

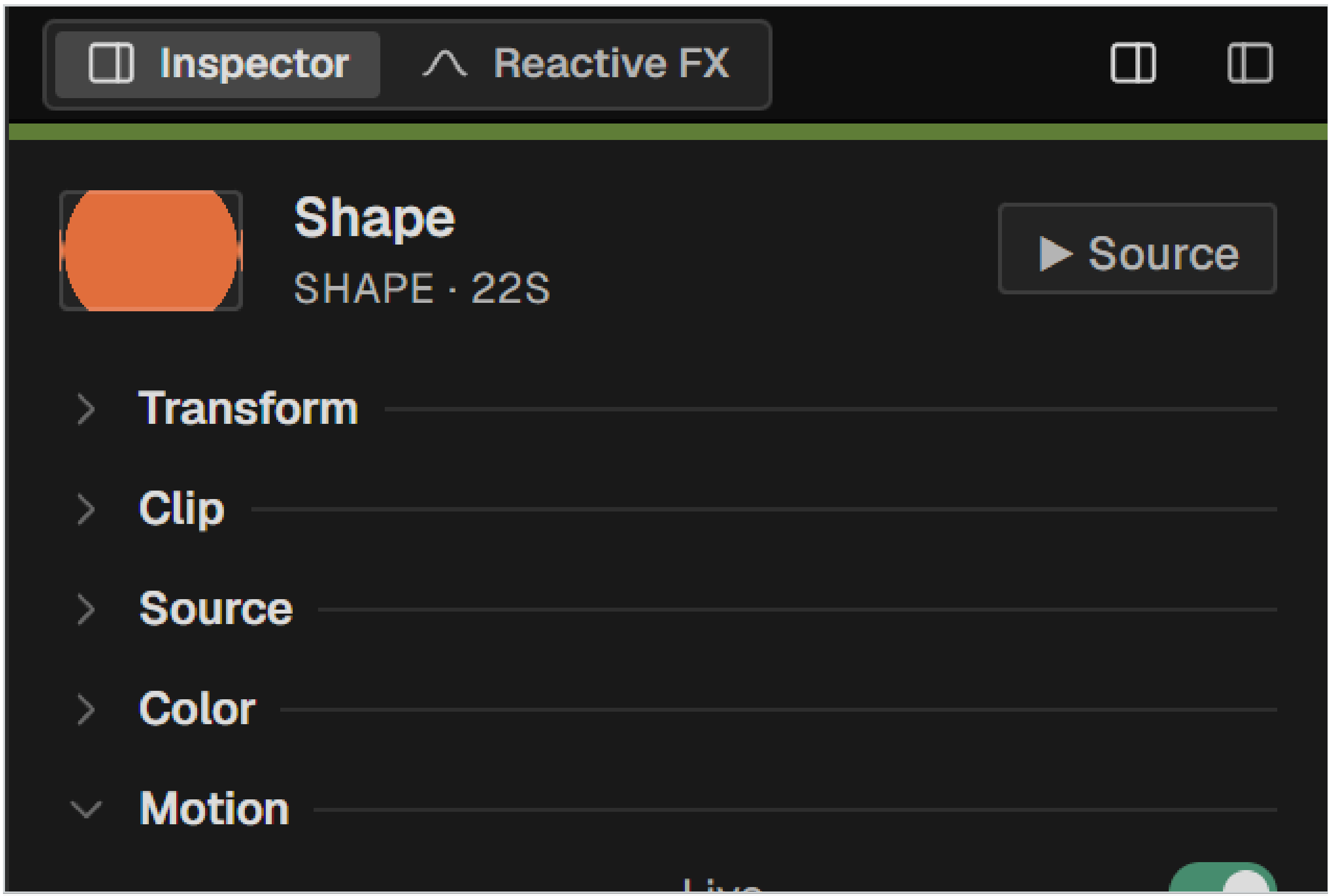
`LUT → Load...` imports a `.cube` 3D LUT and applies it per clip as the final step of the colour chain. This is the intended route for a look developed in a grading application: grade there, export a LUT, apply it here.

> **DOME BRIGHTNESS IS NOT SCREEN BRIGHTNESS**
> A dome is a large, close, curved surface, and material that looks controlled on a monitor is often too bright in the venue. Judge exposure on the dome itself, or at least in the 3D preview, before committing.

Motion, keyframes and automation

Two ways to make something move, meant for different jobs. **Motion** is procedural: endless, parameterised, no keyframes. **Automation** is keyframed: exact values at exact times. They stack, and a mix control balances them.

Motion



The Motion section. Preset chips at the top, then the list of modifiers with their parameter, waveform, speed and mix.

Click a preset chip to add a modifier. Presets are shortcuts, not black boxes — each one stamps ordinary modifiers into the list, and you can retune, disable or delete them individually.

PRESET (DOME)	MOVES
Spin	Rotation about the clip's own centre.
Orbit	Azimuth — travel around the horizon.
Bob	Elevation, back and forth.
Scroll ↕	Elevation, continuously.

Sway	Azimuth, back and forth.
Pulse	Size.
Wobble	Roll.
Flicker	Opacity.
Float	A chord — see below.

In a 2D or room sequence the set is `Rotate`, `Pulse`, `Horizontal`, `Vertical` and `Float`.

Waveforms

Loop
: Grows without end — for continuous travel.

Wave
: Goes and comes back.

Saw
: Rises to the amplitude and is reborn at zero. This is the waveform for a cycle that repeats: an endless scroll, a tunnel. Its `Curve` control from 0 to 100 bends the rise, and an exponential rise is what reads as perspective.

Float

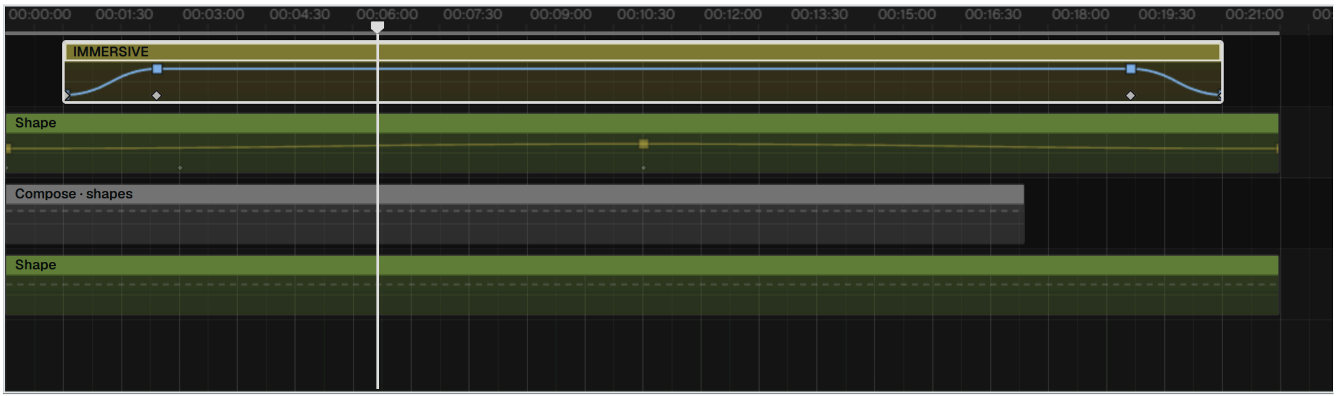
`Float` is the one preset that is a chord rather than a parameter. The sensation of floating does not live in any single axis — it lives in the mixture of horizontal drift, vertical drift and a gentle roll, out of phase with one another. The preset stamps three normal modifiers with deliberately non-multiple periods (11, 14 and 18 seconds) so they never close their cycles together and the movement never visibly repeats. A master `Intensity` control from 0 to 300 % scales the chord while preserving whatever you have retuned by hand.

> **FLOAT MOVES IN THE DOME PLANE**
> In a dome it uses the slide parameters rather than azimuth and elevation, because a degree of azimuth covers far less ground near the zenith than at the horizon — the same drift would otherwise look different depending on where the clip sat.

Keyframes

Every parameter row has a diamond. Click it to set a keyframe at the playhead; click a filled one to remove it. Once a parameter has any keyframe, editing its value writes another one at the playhead rather than moving the whole curve.

The automation lane



Automation mode. Curves are drawn inline over the clips, in the parameter's own colour. The `Auto` button in the editing well turns it on.

Each track shows one parameter's curve at a time, chosen by two chips in the track header: the left one picks the category or device — `Transform`, `Clip`, `Color`, or any Motion or Effect actually applied — and the right one picks the parameter. A ◆ marks whatever is already automated. Touching a parameter in the inspector brings its curve into view automatically, without requiring a keyframe first.

Working on a curve

- Click to add a point, drag to move, right-click to type an exact value.
- Drag a selection over several points, then nudge them, scale them or taper them.
- The shape box gives free transform over a selection.
- Easing presets apply standard cubic-bezier shapes to a segment.
- Simplify thins out a dense curve while keeping its shape.
- Copy a curve and paste it where you click.

Mixing motion and automation

Each Motion modifier has a `Mix` from 0 to 100 %, and the mix itself can be keyframed. That is how you fade a procedural drift in and out over a scene while a keyframed move handles the composition underneath.

Modulation

Beyond keyframes and motion, a parameter can be driven by a stack of modulation layers — LFOs, audio followers and spatial functions — from the modulation panel. It is the most open-ended of the three routes and the one to reach for when you want a value to wander rather than to arrive.

CHAPTER TWENTY

Reactive FX

The second inspector tab: an audio analysis engine and an effects chain whose parameters it can drive. Effects work perfectly well without audio — this is where you wire them to it.

The Reactive FX tab. Audio engine on top, effects chain below.

The audio engine

Source	Which audio drives the analysis.
Band meters	`BASS`, `MID`, `TREB`, `BRT` — live levels for the three frequency bands and overall brightness.
BPM	Detected tempo, or set manually.
Gain	Input level into the analysis.
Gate	Threshold below which nothing responds — keeps room noise from twitching the picture.
Attack / Release	How fast the follower rises and falls, in milliseconds. The engine defaults, which each effect can override.

An effect in the chain

Each entry has a power button, a name, a collapse arrow and a bin, and expands into three groups.

Routing

Band
: Which band drives it — or `None` for a static effect.

Mode
: `Follow` tracks the level; other modes derive from the tempo.

`INV`
: Inverts the response — loud becomes less rather than more.

Response

Intensity	The effect's overall strength. Keyframeable.
Reactivity	How much of that strength the audio controls. At 0 % the effect is static.
Attack / Release	Per-effect override of the engine timings.
Curve	Shapes the response — linear through to sharply exponential.
Bounce	Adds overshoot, so a hit springs past its target and settles.

Parameters

The effect's own controls, each with a keyframe diamond. Chapter 28 lists them for all 26 effects.

> **START WITH REACTIVITY LOW**
> A common mistake is to set reactivity high and then fight the result. Set the look you want statically with `Intensity` first, then raise `Reactivity` until it breathes.

Audio

Audio handling is deliberately modest. The program is a picture editor: it plays a mix, lets you place and trim it, and drives the reactive engine from it.

Audio tracks and clips

Audio tracks live at the bottom of the track column and carry a subtle tint in both the header and the row. Audio clips trim, move and fade like any other, and the waveform is drawn at screen resolution for the visible span only.

The audio inspector

Waveform
A miniature of the clip.

Volume
Clip gain.

Wave scale
Vertical zoom of the waveform display, from ¼× to 8×. This is a *view* preference for the whole timeline — it does not touch the sound or the export.

Fades

Audio fades are the clip's corner handles, the same gesture as video, and the fade *is* the volume. There are no separate fade-in and fade-out fields.

Linked audio and video

Video imported with sound arrives as a linked pair. The two move and trim together; unlink them from the clip context menu when you need to.

Audio in exports

The rule depends on the container:

- **MP4** — stereo audio is embedded in the file.
- **Any other codec** — PNG sequence, HAP — audio is written as a separate file, because those formats either cannot carry it or should not.

Beat markers

A command in the palette detects beats on an audio clip and places locators on them, which then act as snap targets for cutting to music.

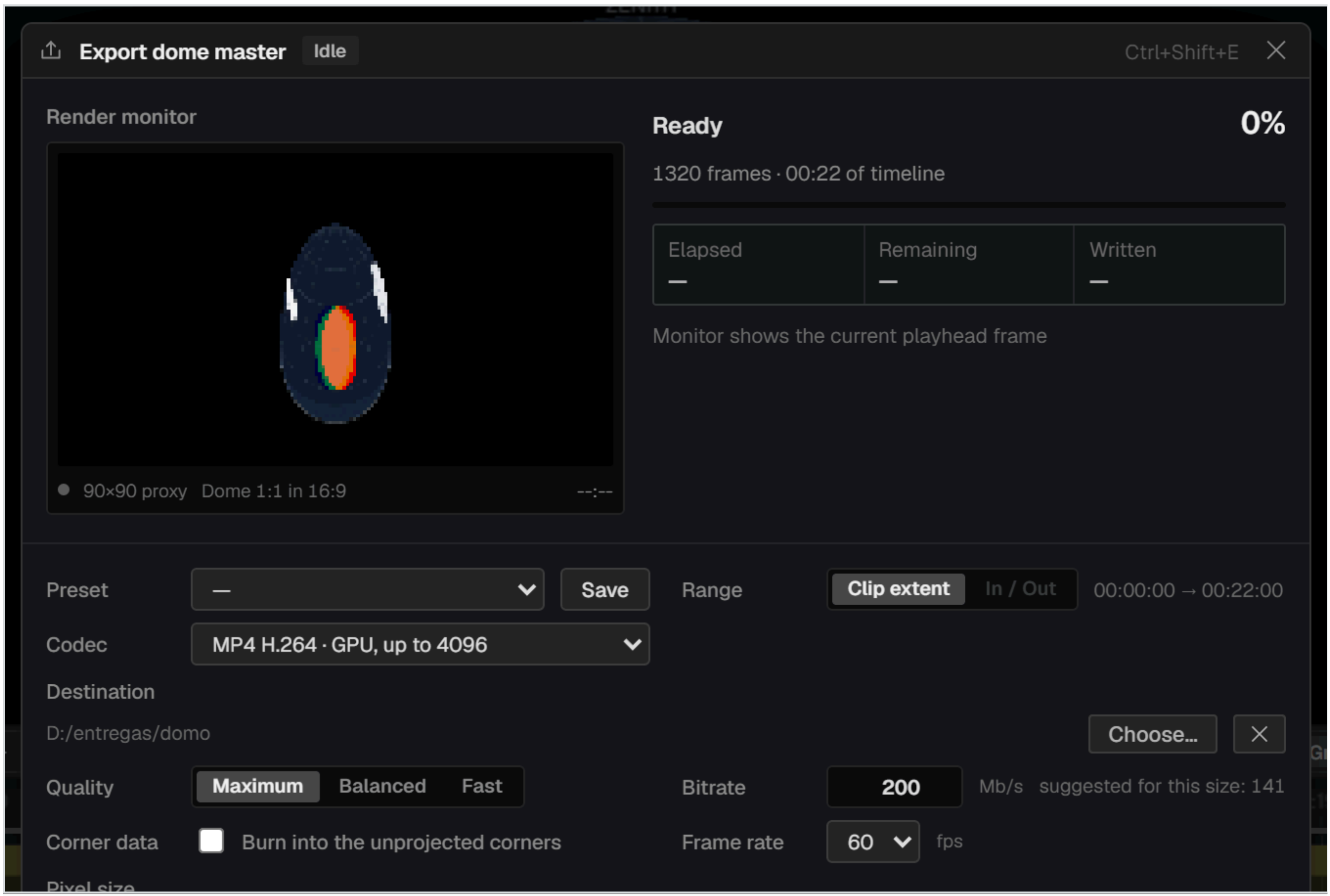
PART FIVE

Delivery

Getting the piece out of the program — as a file, as a burned-in identification, or as a live signal.

CHAPTER TWENTY-TWO

Exporting



The export sheet. Render monitor and status on top; codec, destination, quality and size below. The sheet is draggable and the estimate updates as you change settings.

Before you press Export

1. **Choose the destination folder** with `Choose...`. Doing it first means the render starts when you press the button instead of stopping to ask.
2. **Set the range** — `Clip extent` for everything, or `In / Out` for the marked range.
3. **Pick the codec.**
4. **Check the pixel size** — `Match source` takes it from the sequence.
5. **Read the estimate** at the foot: file size, frame count and dimensions.

The codecs

CODEC	CEILING	USE FOR
PNG sequence · lossless	None	Mastering and archival. Optionally without alpha, on black — still lossless.

MP4 H.264 · GPU	4096 × 4096	The fast path for a 4K dome master.
MP4 H.265 · GPU	8192 × 8192, 10-bit	Large formats, wide room strips, and 10-bit when gradients band.
MP4 · H.264 (built-in)	3072 × 3072	Fallback when the GPU encoder is unavailable.
MP4 · H.265 (built-in)	1080 × 1080	Fallback only. Too small for dome delivery.
MOV · HAP / HAP Q	None	Playback systems that want HAP. Large files, very cheap to decode.
Still frame · PNG	None	A single frame — stills for documentation.

THE TWO H.264 ENTRIES ARE DIFFERENT ENGINES

The entries marked `GPU` use the graphics card's hardware encoder and reach real dome resolutions. The other two use the built-in browser encoder, which refuses anything above 3072 × 3072 for H.264 and about 1080 × 1080 for H.265. For dome work, use the GPU entries.

Quality and bitrate

With a GPU codec selected you get `Quality` — `Maximum`, `Balanced`, `Fast` — and an explicit `Bitrate` in Mb/s, with a suggested figure for the chosen size. For a 4096^2 master, 150–250 Mb/s at `Maximum` is a sensible delivery range.

`Bit depth` offers 10-bit on H.265, which is worth taking for dome work: large smooth gradients across a big surface are exactly where 8-bit banding shows.

Presets

Save a configuration with `Save` next to the `Preset` dropdown and recall it later. Worth doing once per venue.

The render monitor and the queue

The monitor at the top left shows the frame the encoder has just written — not a guess, the actual output. Elapsed, remaining and written counters sit beside it. Jobs run one at a time; a room export in `Each wall + floor` mode queues one job per surface automatically.

Render in place

To bake part of a project rather than deliver it, use `Render in place...` from a clip's context menu — see chapter 15.

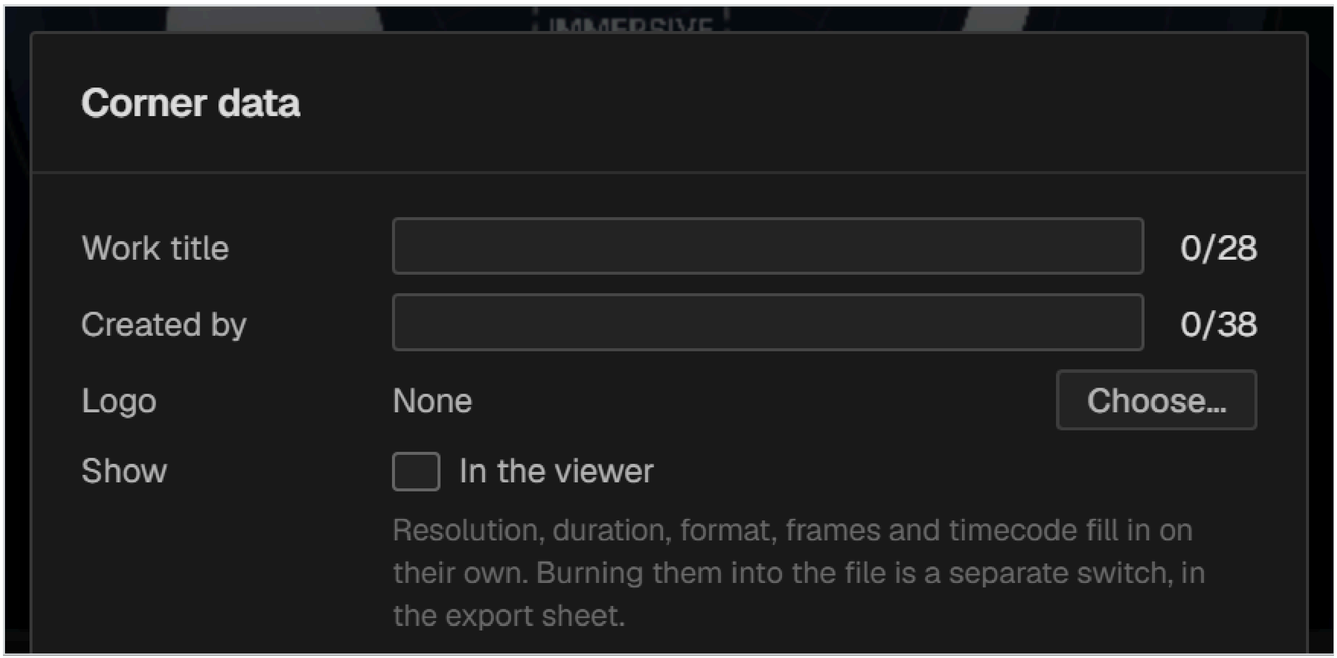
CHECK THE FIRST MINUTE

A 4096^2 export is a long operation. Let it run a minute, stop it, and look at the file in the player you will actually use before committing to the whole piece.

CHAPTER TWENTY-THREE

Corner data

A dome master is a circle inside a square, which leaves four corners that are never projected. Corner data uses them for identification — title, author, running timecode — so a review copy carries its own labelling without touching the picture.



Corner data settings, from `Window → Corner data settings...`

What goes where

Bottom left	The title of the work, in large type.
Bottom right	Running timecode and frame count, advancing with the export.
Top left	A logo image.
Top right	Creator, resolution, duration and format.

Seeing it while you edit

`Window → Show corner data` draws it in the viewer. This is a view setting only.

Recording it into the file

Burning it in is a separate switch, in the export sheet: `Corner data → Burn into the unprojected corners`.

LOOKING IS NOT DELIVERING

The two switches are deliberately separate. Turning the overlay on while you edit never affects a file, and a master you export is clean unless you explicitly asked for corner data to be burned in.

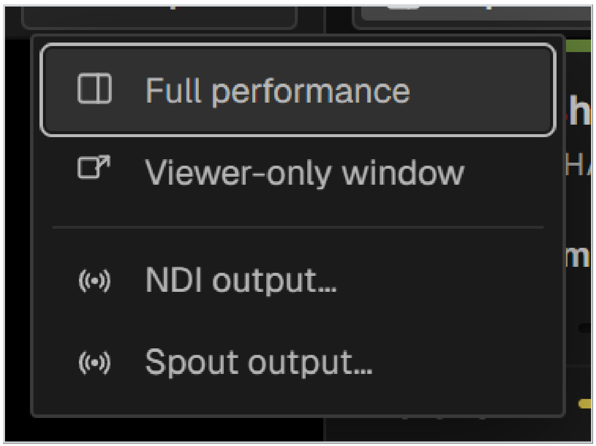
Details worth knowing

- The timecode counts **exported frames**, so it matches the delivered file exactly.
- Text blocks shrink automatically if a long title would collide with the projected circle.
- In a dome sequence the picture is clipped to the circle before the corner data is drawn, so it can never overlap the image.
- An optional black background fills the corners for players that dislike transparency.

CHAPTER TWENTY-FOUR

Live output — NDI and Spout

The composite master can be broadcast live while you work, which is how you drive a dome during a rehearsal or feed a media server without exporting anything.



The Output menu, above the viewer. A pulsing dot marks an active broadcast.

Full performance	The viewer takes over the whole window.
Viewer window	A second window for another display, showing the complementary view.
NDI	Broadcasts the clean master over the network. Any NDI receiver can pick it up.
Spout	Shares the master with another application on the same Windows machine, GPU to GPU with no copy through system memory.

What is broadcast is the **clean master** — no guides, no overlays, no corner data.

NDI input

NDI also works the other way: a live NDI stream can be added as a media item and used as a clip, which is how you bring in a live camera or another machine's output.

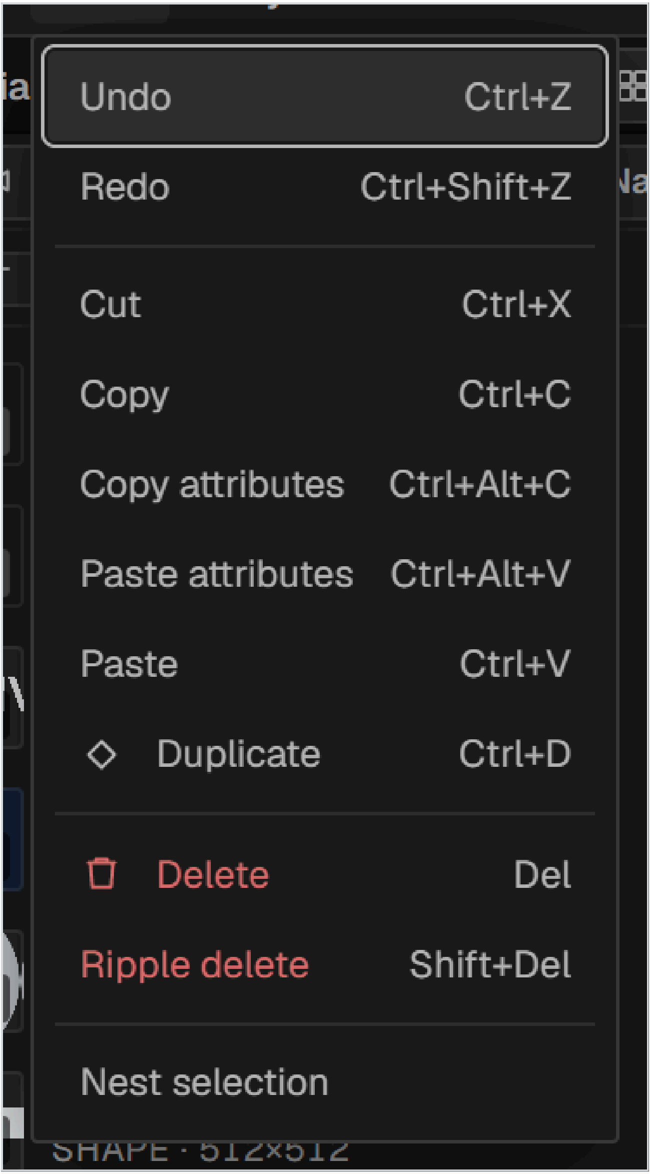
> **NDI NEEDS ITS RUNTIME**
> If the NDI runtime is not installed the program offers a link to it. Spout is Windows-only.

PART SIX

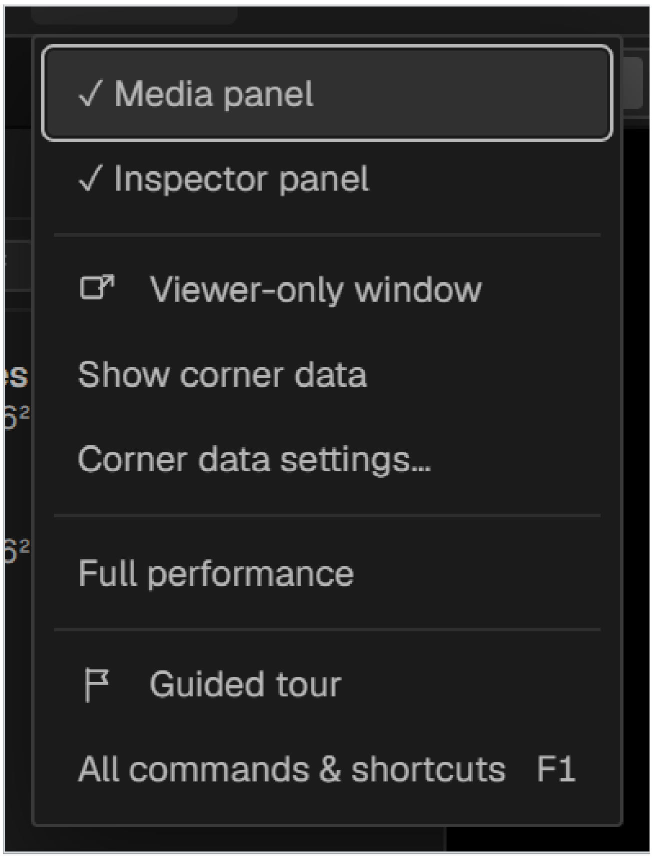
Reference

Tables to look things up in.

Menu reference



The Edit menu.



The Window menu.

File

New project...	Ctrl + N	Goes to the start screen. The open project stays behind it.
Open...	Ctrl + O	Opens .isp, and legacy .ise / .rdome.
Save	Ctrl + S	
Save As...	Ctrl + Shift + S	
Export...	Ctrl + Shift + E	

Edit

Undo / Redo	Ctrl + Z · Ctrl + Shift + Z	
Cut	Ctrl + X	
Copy / Paste	Ctrl + C · Ctrl + V	Operates on the whole selection.
Copy attributes	Ctrl + Alt + C	Chapter 11.
Paste attributes	Ctrl + Alt + V	
Duplicate	Ctrl + D	
Delete	Del	Leaves a gap.
Ripple delete	Shift + Del	Closes the gap.
Nest selection		Chapter 15.

Project

Sequence settings...		Resolution, frame rate, dome coverage.
Dome coverage (FOV)...		Dome sequences only.
Room geometry...		Room sequences only. Reconfigures; does not create a new project.
New sequence...		Dome, 2D or 360 Room.
Preferences...	Ctrl + ,	

Window

Media panel		Show or hide.
Inspector panel		Show or hide.
Viewer-only window		Second window, complementary view.
Show corner data		Draws it in the viewer. Does not affect files.
Corner data settings...		Fills in the title, author and logo.
Full performance		Viewer fills the window.
Guided tour		Relaunches the onboarding tour.
All commands & shortcuts	F1	The command palette.

CHAPTER TWENTY-SIX

Keyboard shortcuts

On macOS read Ctrl as ⌘. The complete list is always available in the program itself with F1.

Transport

Space	Play / Pause
Home	Go to start
End	Go to end
J	Shuttle back (J · press again = 2×/4×/8×)
L	Shuttle forward (L · press again = 2×/4×/8×)
K	Stop shuttle

File

Ctrl + N	New project
Ctrl + S	Save
Ctrl + I	Import media...
Ctrl + Shift + E	Export master...
Ctrl + O	Open project
Ctrl + ,	Preferences...
Ctrl + Shift + S	Save As... (new file)

Edit

Ctrl + Z	Undo
Ctrl + Shift + Z	Redo
Ctrl + D	Duplicate
Del	Delete
Shift + Del	Ripple delete
Ctrl + C	Copy
Ctrl + V	Paste

Timeline

Ctrl + E	Split at selection / playhead
Alt + ←/→	Nudge clip ±1 frame / ±1 s
+	Zoom in
–	Zoom out
↑	Previous cut
↓	Next cut

Locators

M	Add locator
.	Next
,	Previous

Tools

V	Select
H	Hand / Pan
B	Razor
Z	Zoom

CHAPTER TWENTY-SEVEN

Compose reference

Dome types

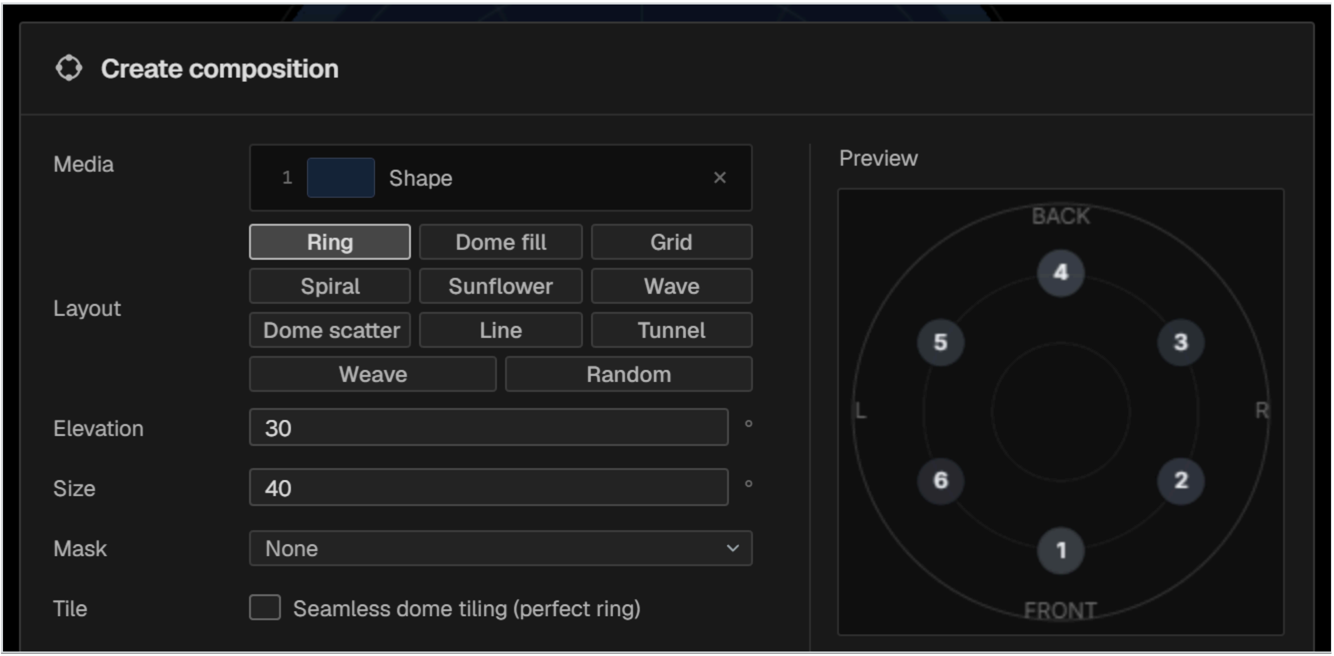
TYPE	PRINCIPAL PARAMETERS
Ring	Count, elevation, size, ring radius.
Dome fill	Rings, segments, ring gap, segment gap, offset, tiles.
Spiral	Count, turns, size.
Sunflower	Count, size.
Wave	Count, amplitude, frequency.
Dome scatter	Count, size, jitter.
Random	Count, size, jitter.
Tunnel	Count, from, to, speed, depth, spin, helix, fade in, fade out.

Flat types

TYPE	PRINCIPAL PARAMETERS
Grid	Columns, rows, gap.
Line	Count, spacing, angle.
Row / Column	Count, spacing.
Weave	Layout, strips, strip width, packing, long side, direction, speed per family, order.

Common to all

Media	The basket — which sources, and in what order.
Mask	Shape and size (0–100 %) applied to every element.
Order	Shuffle which source goes where.



Ring — the simplest type, and a good place to learn the dialog.

CHAPTER TWENTY-EIGHT

Effects reference

Twenty-six effects, grouped by family. Every parameter can be keyframed, and every effect can be driven by audio from the Reactive FX tab. Effects marked **feedback** read the previous frame, so their look builds up over time.

Distortion

EFFECT	PARAMETERS
Scale Pulse	Amount
Rotate	Angle · Spin
Glitch	Blocks · RGB · Rate
Datamosh · feedback	Smear · Dir X · Dir Y
Slice	Slices · Rate
Pixelate	Blocks
Kaleidoscope	Segments · Angle · Zoom
Mirror	Mode 0-4
Wave	Amount · Frequency · Speed
Zoom Blur	Strength · Center X · Center Y
Noise Warp	Scale · Amount · Speed

Stylize

EFFECT	PARAMETERS
Strobe	Flash · Hardness
Edge	Thickness · Mix
Posterize	Levels
Scanlines / CRT	Lines · Strength · Chroma
Bloom / Glow	Threshold · Radius · Boost

Colour

EFFECT	PARAMETERS
RGB Split	Angle · Spread
Hue Shift	Shift
Chroma Pulse	Strength · Center X · Center Y
Flash	Mode W/B/Inv

Feedback

EFFECT	PARAMETERS
Trails / Echo · feedback	Decay
Feedback Zoom · feedback	Zoom · Rotate · Decay
Feedback Flow · feedback	Zoom · Rotate · Hue Cycle · Warp · Decay

Dome

EFFECT	PARAMETERS
Dome Rings	Rings · Sharpness · Travel
Spiral Twist	Twist · Rotation
Tunnel	Depth · Flight

CHAPTER TWENTY-NINE

Performance and troubleshooting

Making playback smooth

1. **Generate proxies for heavy footage.** Right-click the media item. With long-GOP camera masters this is the single largest gain available.
2. **Drop the viewer quality** to ½ or ¼. It does not affect the export.
3. **Cache heavy compositions** with a composition proxy, so playback decodes one stream instead of many.
4. **Render in place** anything that is finished and expensive.
5. **Hide what you are not using.** Mute tracks you are not judging.

Common situations

Something looks soft inside a composition

Compositing passes through an intermediate buffer, and its resolution is chosen from the output size. Export at the sequence's real resolution rather than a smaller one, and the difference largely disappears.

The picture spills outside the dome circle

It does not — anything outside the inscribed circle is not part of a fulldome master and will be black, or transparent with alpha. Reduce the clip's size or move it.

An export refuses a resolution

Check which codec entry you chose. The built-in H.264 stops at 3072 × 3072 and the built-in H.265 at about 1080 × 1080; the entries marked `GPU` reach 4096 and 8192.

A clip plays every second frame

Its proxy was generated at the wrong frame rate because the source's rate could not be detected. Delete the proxy and generate it again.

Turning off a loop shortened the clip

Switching looping off trims the clip to what remains of the file. Use `From clip` to redefine a cycle instead, and undo to recover.

A project opened with the wrong format or framing

Factory defaults are applied before every project is read, so nothing is inherited from the previous one. If a sequence still looks wrong, check `Project → Sequence settings…`.

The 3D view is black

On a hybrid-graphics laptop, make sure the program is running on the discrete GPU. It requests this automatically; a system-level override can defeat it.

Recovering work

- Projects save atomically with a `.bak` alongside.
- `Restore last autosave` and `Recovery history…` are in the command palette.
- `Save incremental` writes `_v01`, `_v02`… — worth doing before any large restructuring.

Reporting a problem

`Save diagnostics log…` in the command palette writes a log file describing the environment and recent activity. Send it with a description of what you were doing.

Colophon

Immersive Studio Pro — user manual for version 1.0.

Created by Alma Digital Studio. Creative direction by Beltrán.

Every screenshot in this manual was captured from the running application, and every menu, shortcut, effect and parameter table was read from the program itself rather than transcribed by hand.

August 2026.